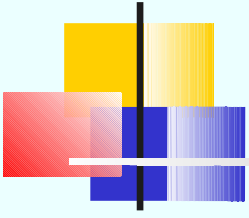


Programming with Android

Android intermediate

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Posts & Telecommunications Institute of Technology (PTIT)



Content

- Custom components
- Drag & drop
- Notifications
- Location-based service
- Sending email/SMS
- Phone calling
- Alert dialog
- Animation
- Audio/bluetooth/camera

References



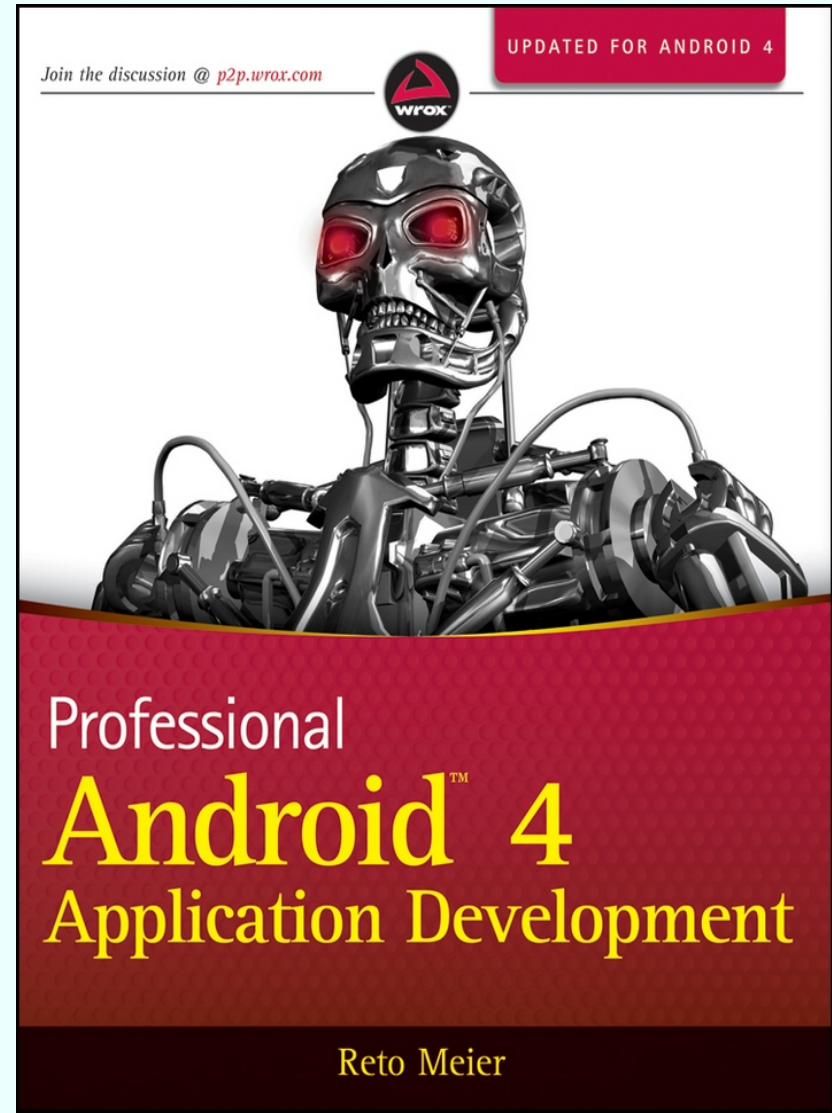
www.tutorialspoint.com

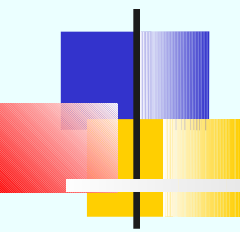


<https://www.facebook.com/tutorialspointindia>

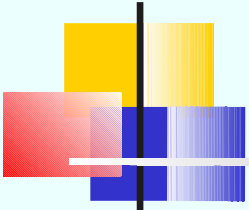


<https://twitter.com/tutorialspoint>





Custom components



Custom components

Customize component by

- Extend from an existent component
- Combine some components and pack them into a single object which could be used as a component



Extend from a widget

Define DateView from TextView:

```
public class DateView extends TextView {  
    public DateView(Context context) {  
        super(context);  
        //--- Additional custom code ---  
    }  
  
    public DateView(Context context, AttributeSet attrs) {  
        super(context, attrs);  
        //--- Additional custom code ---  
    }  
  
    public DateView(Context context, AttributeSet attrs, int defStyle)  
        super(context, attrs, defStyle);  
        //--- Additional custom code ---  
    }  
}
```



Using in code

In java code:

```
@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    DateView dateView = new DateView(this);
    setContentView(dateView);
}
```



Using in layout xml

In layout xml file:

```
<RelativeLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:paddingBottom="@dimen/activity_vertical_margin"
android:paddingLeft="@dimen/activity_horizontal_margin"
android:paddingRight="@dimen/activity_horizontal_margin"
android:paddingTop="@dimen/activity_vertical_margin"
tools:context=".MainActivity" >
```

```
    <com.example.dateviewdemo.DateView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:textColor="#fff"
        android:textSize="40sp"
        android:background="#000"
    />
```

```
</RelativeLayout>
```



Custom attributes (1)

If the objective object is:

```
<com.example.dateviewdemo.DateView  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:textColor="#fff"  
    android:textSize="40sp"  
    custom:delimiter="-"  
    custom:fancyText="true"  
/>
```



Custom attributes (2)

Step 1: add/create res/values/attrs.xml

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <declare-styleable name="DateView">
        <attr name="delimiter" format="string"/>
        <attr name="fancyText" format="boolean"/>
    </declare-styleable>
</resources>
```



Custom attributes (3)

Step 2: read xml file to component

```
TypedArray a = context.obtainStyledAttributes(attrs, R.styleable.DateView);
final int N = a.getIndexCount();
for (int i = 0; i < N; ++i){
    int attr = a.getIndex(i);
    switch (attr){
        case R.styleable.DateView_delimiter:
            String delimiter = a.getString(attr);
            //...do something with delimiter...
            break;
        case R.styleable.DateView_fancyText:
            boolean fancyText = a.getBoolean(attr, false);
            //...do something with fancyText...
            break;
    }
}
a.recycle();
```



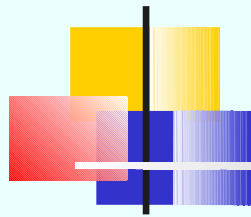
Custom attributes (4)

Step 3: using attributes to define component in layout xml

```
<RelativeLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:tools="http://schemas.android.com/tools"
xmlns:custom="http://schemas.android.com/apk/res/com.example.dateviewdemo"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:paddingBottom="@dimen/activity_vertical_margin"
android:paddingLeft="@dimen/activity_horizontal_margin"
android:paddingRight="@dimen/activity_horizontal_margin"
android:paddingTop="@dimen/activity_vertical_margin"
tools:context=".MainActivity" >
```

```
    <com.example.dateviewdemo.DateView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:textColor="#fff"
        android:textSize="40sp"
        custom:delimiter="-"
        custom:fancyText="true"
    />
```

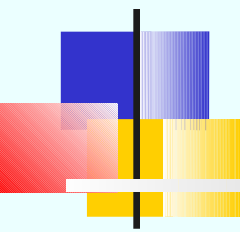
```
</RelativeLayout>
```



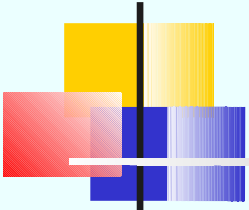
Exercises

Create a component to login:

- There are two TextView, two EditText, a Button
- Can change the text of the two textviews
- Can get an object of (username, password) from that component
- Handle login click



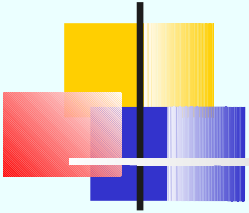
Drag & drop



Drag/drop process

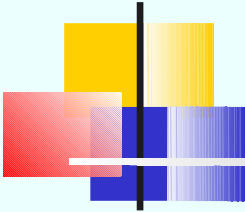
There are 4 steps:

- Started: occurs when user starts dragging an item in a layout
- Continuing: The user continues the drag
- Dropped: The user releases the dragged item within the bounding box of a View.
- Ended: Just after the action type `ACTION_DROP`, the system sends out a drag event with action type `ACTION_DRAG_ENDED` to indicate that the drag operation is over.



Drag/drop events

- ACTION_DRAG_STARTED
- ACTION_DRAG_ENTERED
- ACTION_DRAG_LOCATION
- ACTION_DRAG_EXITED
- ACTION_DROP
- ACTION_DRAG_ENDED



Drag/drop methods

- `getAction()`
- `getClipData()`
- `getClipDescription()`
- `getResult()`
- `getX()`
- `getY()`
- `toString()`



Example (1)

MainActivity.java (1):

```
package com.example.dragndropdemo;
import android.os.Bundle;
import android.app.Activity;
import android.content.ClipData;
import android.content.ClipDescription;
import android.util.Log;
import android.view.DragEvent;
import android.view.View;
import android.view.View.DragShadowBuilder;
import android.view.View.OnDragListener;
import android.widget.*;

public class MainActivity extends Activity{
    ImageView ima;
    private static final String IMAGEVIEW_TAG = "Android Logo";
    String msg;
    private android.widget.RelativeLayout.LayoutParams layoutParams;

    @Override
    public void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        ima = (ImageView)findViewById(R.id.iv_logo);
        // Sets the tag
        ima.setTag(IMAGEVIEW_TAG);
    }
}
```



Example (2)

MainActivity.java (2):

```
ima.setOnLongClickListener(new View.OnLongClickListener() {
    @Override
    public boolean onLongClick(View v) {
        ClipData.Item item = new
        ClipData.Item((CharSequence)v.getTag());
        String[] mimeTypes = {ClipDescription.MIMETYPE_TEXT_PLAIN};
        ClipData dragData = new ClipData(v.getTag().toString(),
            mimeTypes, item);
        // Instantiates the drag shadow builder.
        View.DragShadowBuilder myShadow = new DragShadowBuilder(ima);
        // Starts the drag
        v.startDrag(dragData, // the data to be dragged
            myShadow, // the drag shadow builder
            null, // no need to use local data
            0 // flags (not currently used, set to 0)
        );
        return true;
    }
});
```



Example (3)

MainActivity.java (3):

```
// Create and set the drag event listener for the View
ima.setOnDragListener( new OnDragListener(){
    @Override
    public boolean onDrag(View v, DragEvent event){
        switch(event.getAction()){
            case DragEvent.ACTION_DRAG_STARTED:
                layoutParams = (RelativeLayout.LayoutParams)
                    v.getLayoutParams();
                Log.d(msg, "Action is DragEvent.ACTION_DRAG_STARTED");
                // Do nothing
                Break;
            case DragEvent.ACTION_DRAG_ENTERED:
                Log.d(msg, "Action is DragEvent.ACTION_DRAG_ENTERED");
                int x_cord=(int)event.getX();
                int y_cord=(int)event.getY();
                break;
            case DragEvent.ACTION_DRAG_EXITED :
                Log.d(msg, "Action is DragEvent.ACTION_DRAG_EXITED");
                x_cord = (int) event.getX();
                y_cord = (int)event.getY();
                layoutParams.leftMargin = x_cord;
                layoutParams.topMargin = y_cord;
                v.setLayoutParams(layoutParams);
                break;
```



Example (4)

MainActivity.java (4):

```
        case DragEvent.ACTION_DRAG_LOCATION :
            Log.d(msg, "Action is DragEvent.ACTION_DRAG_LOCATION");
            x_cord = (int) event.getX();
            y_cord = (int) event.getY();
            break;
        case DragEvent.ACTION_DRAG_ENDED :
            Log.d(msg, "Action is DragEvent.ACTION_DRAG_ENDED");
            // Do nothing
            break;
        case DragEvent.ACTION_DROP:
            Log.d(msg, "ACTION_DROP event");
            // Do nothing
            break;
        default: break;
    }
    return true;
}
    });
}
```



Example (5)

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
xmlns:android="http://schemas.android.com/apk/res/android"
android:id="@+id/container"
android:layout_width="fill_parent"
android:layout_height="fill_parent"
android:orientation="vertical" >

    <ImageView
        android:id="@+id/iv_logo"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:src="@drawable/logo"
        android:contentDescription="@string/drag_drop" />
</RelativeLayout>
```



Example (6)

strings.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <string name="app_name">DragNDropDemo</string>
    <string name="action_settings">Settings</string>
    <string name="hello_world">Hello world!</string>
    <string name="drag_drop">Click on the image to drag and drop</string>
</resources>
```



Example (7)

AndroidManifest.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
package="com.example.guidemo"
android:versionCode="1"
android:versionName="1.0" >

    <uses-sdk
        android:minSdkVersion="16"
        android:targetSdkVersion="17" />

    <application
        android:allowBackup="true"
        android:icon="@drawable/ic_launcher"
        android:label="@string/app_name"
        android:theme="@style/AppTheme" >

        <activity
            android:name="com.example.guidemo.MainActivity"
            android:label="@string/app_name" >
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>
</manifest>
```

Example (8)

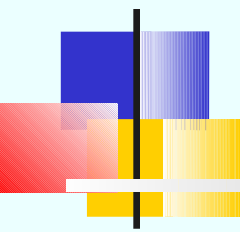


Exercises

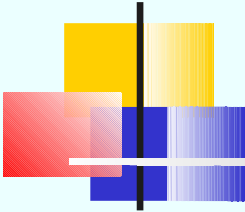


Game remember photo:

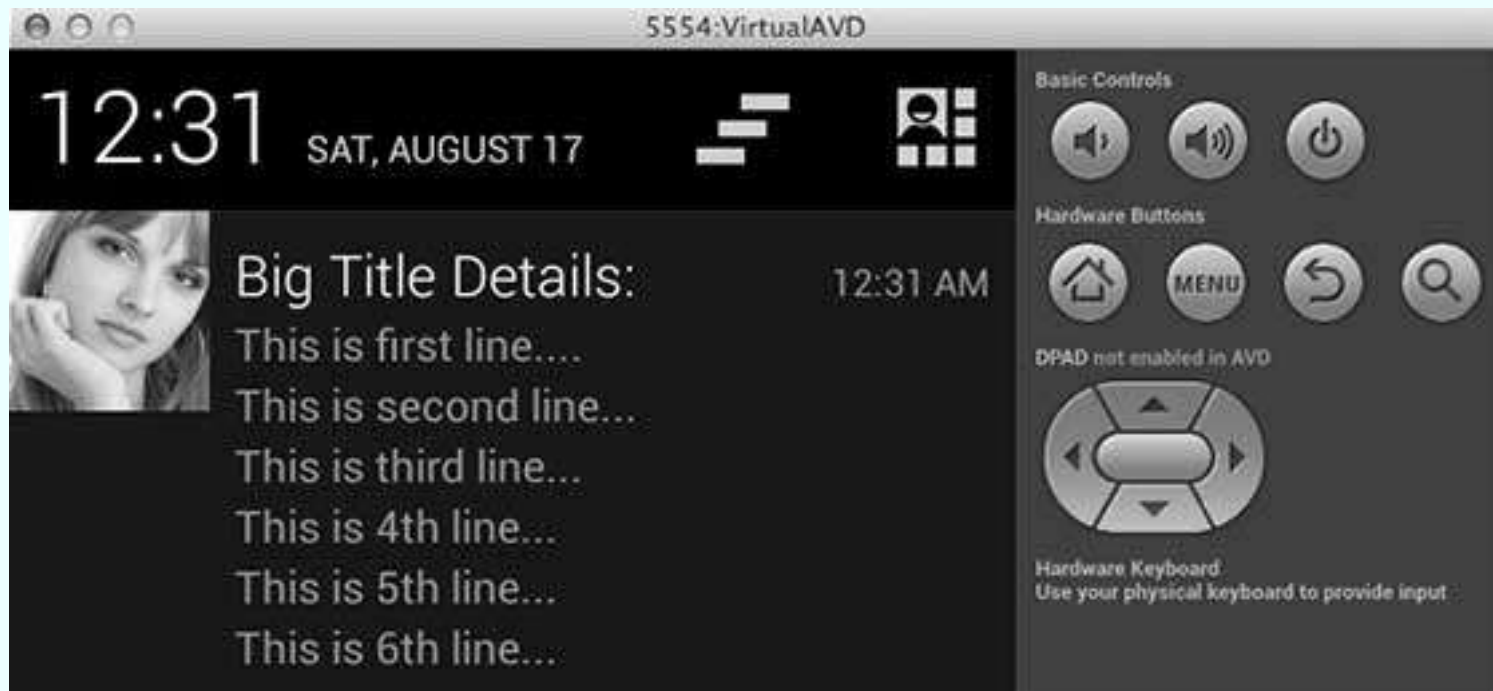
- A photo is cut into several pices
- The inteface has three zones: original photo, set of pices, and a zone to re-construct the photo
- Enable to drag a pice of photo and to drop into zone of re-construction, at a position, rotage the pice
- Verify if a pice is in correct position
- Verify if the photo is done
- Count the time to solve
- Save the best time for each player

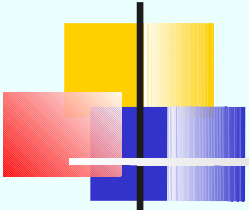


Notifications



Notifications





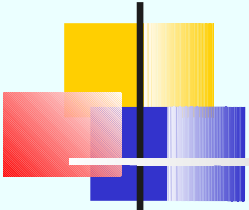
Create notification (1)

Step 1 – create notification builder:

```
NotificationCompat.Builder mBuilder = new  
    NotificationCompat.Builder(this)
```

Step 2 – set properties

```
mBuilder.setSmallIcon(R.drawable.notification_icon);  
mBuilder.setContentTitle("Notification Alert, Click Me!");  
mBuilder.setContentText("Hi, This is Android Notification Detail!");
```



Create notification (2)

Step 3 – attach actions:

```
Intent resultIntent = new Intent(this, ResultActivity.class);
TaskStackBuilder stackBuilder = TaskStackBuilder.create(this);
stackBuilder.addParentStack(ResultActivity.class);

// Adds the Intent that starts the Activity to the top of the stack
stackBuilder.addNextIntent(resultIntent);
PendingIntent resultPendingIntent =
    stackBuilder.getPendingIntent(0, PendingIntent.FLAG_UPDATE_CURRENT);
mBuilder.setContentIntent(resultPendingIntent);
```

Step 4 – issue the notification

```
NotificationManager mNotificationManager =
    (NotificationManager) getSystemService(Context.NOTIFICATION_SERVICE);
// notificationID allows you to update the notification later on.
mNotificationManager.notify(notificationID, mBuilder.build());
```



Example (1)

MainActivity.java (1):

```
package com.example.notificationdemo;
import android.os.Bundle;
import android.app.Activity;
import android.app.NotificationManager;
import android.app.PendingIntent;
import android.app.TaskStackBuilder;
import android.content.Context;
import android.content.Intent;
import android.support.v4.app.NotificationCompat;
import android.util.Log;
import android.view.View;
import android.widget.Button;

public class MainActivity extends Activity {
    private NotificationManager mNotificationManager;
    private int notificationID = 100;
    private int numMessages = 0;

    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        Button startBtn = (Button) findViewById(R.id.start);
        startBtn.setOnClickListener(new View.OnClickListener() {
            public void onClick(View view) {
                displayNotification();
            }
        });
    }
}
```



Example (2)

MainActivity.java (2):

```
Button cancelBtn = (Button) findViewById(R.id.cancel);
cancelBtn.setOnClickListener(new View.OnClickListener() {
    public void onClick(View view) {
        cancelNotification();
    }
});
Button updateBtn = (Button) findViewById(R.id.update);
updateBtn.setOnClickListener(new View.OnClickListener() {
    public void onClick(View view) {
        updateNotification();
    }
});
protected void cancelNotification() {
    Log.i("Cancel", "notification");
    mNotificationManager.cancel(notificationID);
}
```



Example (3)

MainActivity.java (3):

```
protected void displayNotification() {
    Log.i("Start", "notification");
    /* Invoking the default notification service */
    Builder mBuilder = new NotificationCompat.Builder(this);
    mBuilder.setContentTitle("New Message");
    mBuilder.setContentText("You've received new message.");
    mBuilder.setTicker("New Message Alert!");
    mBuilder.setSmallIcon(R.drawable.woman);

    //Increase notification number every time a new notification arrives
    mBuilder.setNumber(++numMessages);

    /* Creates an explicit intent for an Activity in your app */
    Intent resultIntent = new Intent(this, NotificationView.class);
    TaskStackBuilder stackBuilder = TaskStackBuilder.create(this);
    stackBuilder.addParentStack(NotificationView.class);

    //Adds the Intent that starts the Activity to the top of the stack
    stackBuilder.addNextIntent(resultIntent);
    PendingIntent resultPendingIntent = stackBuilder.getPendingIntent(
        0, PendingIntent.FLAG_UPDATE_CURRENT);
    mBuilder.setContentIntent(resultPendingIntent);
    mNotificationManager = (NotificationManager)
        getSystemService(Context.NOTIFICATION_SERVICE);

    //notificationID allows you to update the notification later on.
    mNotificationManager.notify(notificationID, mBuilder.build());
}
```



Example (4)

MainActivity.java (4):

```
protected void updateNotification() {
    Log.i("Update", "notification");
    /* Invoking the default notification service */
    NotificationCompat.Builder mBuilder =
        new NotificationCompat.Builder(this);
    mBuilder.setContentTitle("Updated Message");
    mBuilder.setContentText("You've got updated message.");
    mBuilder.setTicker("Updated Message Alert!");
    mBuilder.setSmallIcon(R.drawable.woman);

    //Increase notification number every time a new notification arrives
    mBuilder.setNumber(++numMessages);

    /* Creates an explicit intent for an Activity in your app */
    Intent resultIntent = new Intent(this, NotificationView.class);
    TaskStackBuilder stackBuilder = TaskStackBuilder.create(this);
    stackBuilder.addParentStack(NotificationView.class);

    //Adds the Intent that starts the Activity to the top of the stack
    stackBuilder.addNextIntent(resultIntent);
    PendingIntent resultPendingIntent = stackBuilder.getPendingIntent(
        0, PendingIntent.FLAG_UPDATE_CURRENT);
    mBuilder.setContentIntent(resultPendingIntent);
    mNotificationManager = (NotificationManager)
        getSystemService(Context.NOTIFICATION_SERVICE);

    /* Update the existing notification using same notification ID */
    mNotificationManager.notify(notificationID, mBuilder.build());
}
```



Example (5)

NotificationView.java:

```
package com.example.notificationdemo;
import android.os.Bundle;
import android.app.Activity;

public class NotificationView extends Activity{

    @Override
    public void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.notification);
    }
}
```



Example (6)

activity_main.xml:

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="fill_parent"
    android:layout_height="fill_parent"
    android:orientation="vertical" >
```

```
    <Button android:id="@+id/start"
        android:layout_width="fill_parent"
        android:layout_height="wrap_content"
        android:text="@string/start_note"/>
```

```
    <Button android:id="@+id/cancel"
        android:layout_width="fill_parent"
        android:layout_height="wrap_content"
        android:text="@string/cancel_note" />
```

```
    <Button android:id="@+id/update"
        android:layout_width="fill_parent"
        android:layout_height="wrap_content"
        android:text="@string/update_note" />
```

```
</LinearLayout>
```



Example (7)

res/layout/notification.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:layout_width="fill_parent"
    android:layout_height="fill_parent" >

    <TextView
        android:layout_width="fill_parent"
        android:layout_height="400dp"
        android:text="Hi, Your Detailed notification view goes here...."
    </LinearLayout>
```



Example (8)

res/layout/strings.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <string name="app_name">NotificationDemo</string>
    <string name="action_settings">Settings</string>
    <string name="hello_world">Hello world!</string>
    <string name="start_note">Start Notification</string>
    <string name="cancel_note">Cancel Notification</string>
    <string name="update_note">Update Notification</string>
</resources>
```

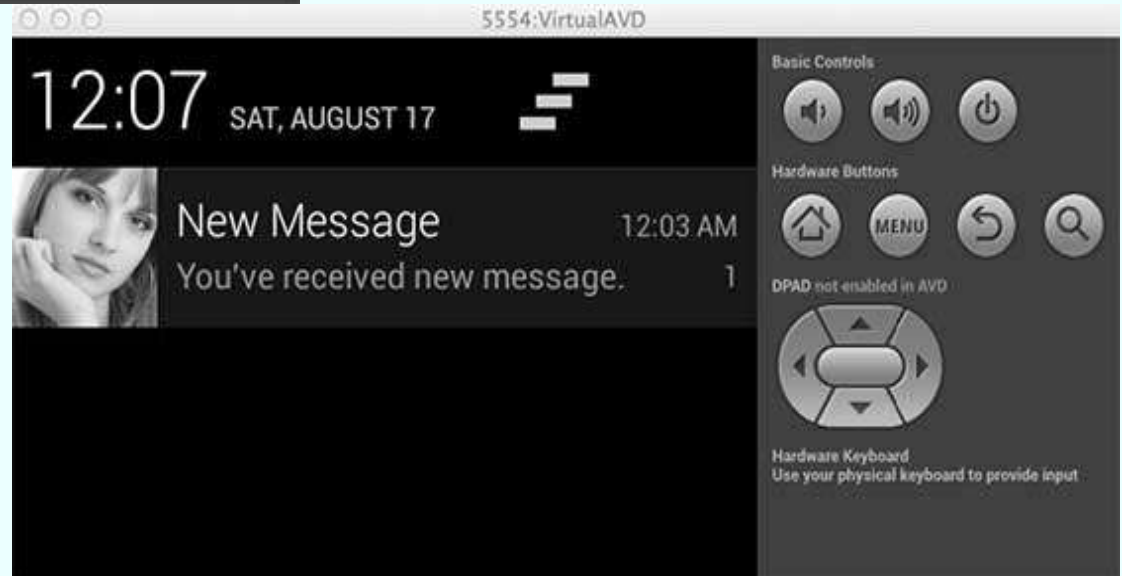


Example (9)

AndroidManifest.xml:

```
<application
    android:allowBackup="true"
    android:icon="@drawable/ic_launcher"
    android:label="@string/app_name"
    android:theme="@style/AppTheme" >
    <activity
        android:name="com.example.notificationdemo.MainActivity"
        android:label="@string/app_name" >
        <intent-filter>
            <action android:name="android.intent.action.MAIN" />
            <category android:name="android.intent.category.LAUNCHER" />
        </intent-filter>
    </activity>
    <activity android:name=".NotificationView"
        android:label="Details of notification"
        android:parentActivityName=".MainActivity">
        <meta-data
            android:name="android.support.PARENT_ACTIVITY"
            android:value=".MainActivity"/>
    </activity>
</application>
```

Example (10)





Big view notification (1)

Change displayNotification() (1):

```
protected void displayNotification() {
    Log.i("Start", "notification");
    /* Invoking the default notification service */
    NotificationCompat.Builder mBuilder =
        new NotificationCompat.Builder(this);
    mBuilder.setContentTitle("New Message");
    mBuilder.setContentText("You've received new message.");
    mBuilder.setTicker("New Message Alert!");
    mBuilder.setSmallIcon(R.drawable.woman);

    //Increase notification number every time a new notification arrives
    mBuilder.setNumber(++numMessages);

    /* Add Big View Specific Configuration */
    NotificationCompat.InboxStyle inboxStyle =
        new NotificationCompat.InboxStyle();
    String[] events = new String[6];
    events[0] = new String("This is first line....");
    events[1] = new String("This is second line...");
    events[2] = new String("This is third line...");
    events[3] = new String("This is 4th line...");
    events[4] = new String("This is 5th line...");
    events[5] = new String("This is 6th line...");
}
```



Big view notification (2)

Change displayNotification() (2):

```
// Sets a title for the Inbox style big view
inboxStyle.setBigContentTitle("Big Title Details:");

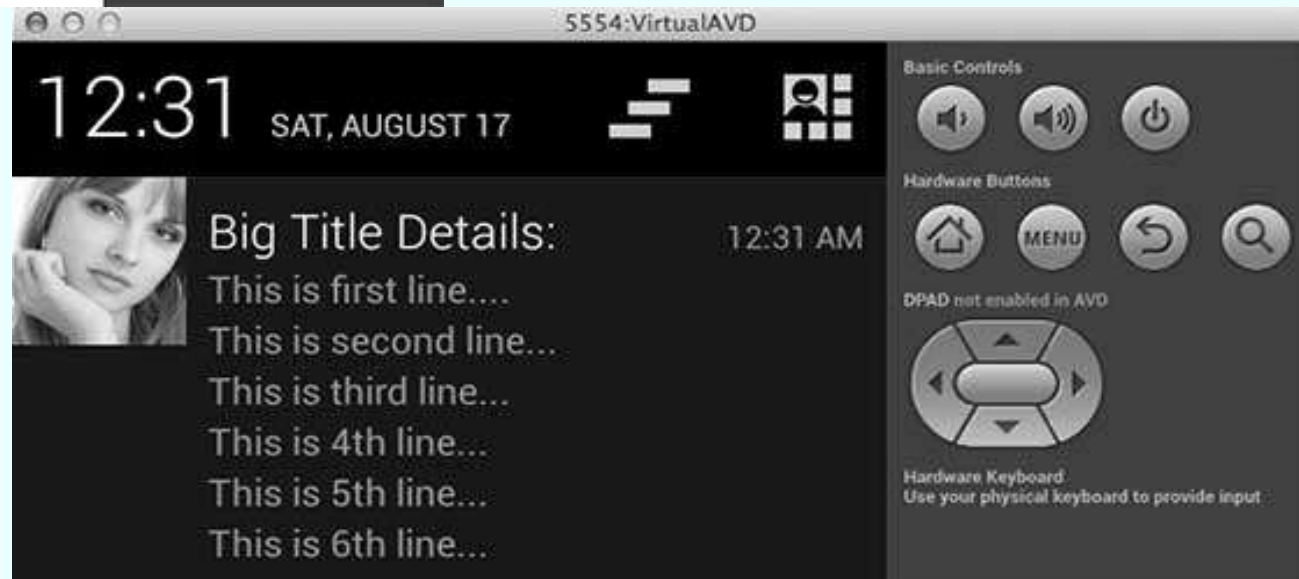
// Moves events into the big view
for (int i=0; i < events.length; i++) {
    inboxStyle.addLine(events[i]);
}
mBuilder.setStyle(inboxStyle);

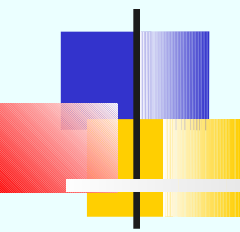
/* Creates an explicit intent for an Activity in your app */
Intent resultIntent = new Intent(this, NotificationView.class);
TaskStackBuilder stackBuilder = TaskStackBuilder.create(this);
stackBuilder.addParentStack(NotificationView.class);

//Adds the Intent that starts the Activity to the top of the stack
stackBuilder.addNextIntent(resultIntent);
PendingIntent resultPendingIntent = stackBuilder.getPendingIntent(
    0, PendingIntent.FLAG_UPDATE_CURRENT);
mBuilder.setContentIntent(resultPendingIntent);
MNotificationManager = (NotificationManager)
    getSystemService(Context.NOTIFICATION_SERVICE);

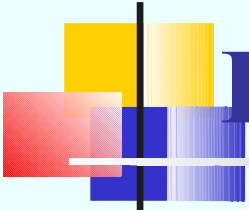
//notificationID allows you to update the notification later on.
mNotificationManager.notify(notificationID, mBuilder.build());
}
```

Example (10)





Location-based service



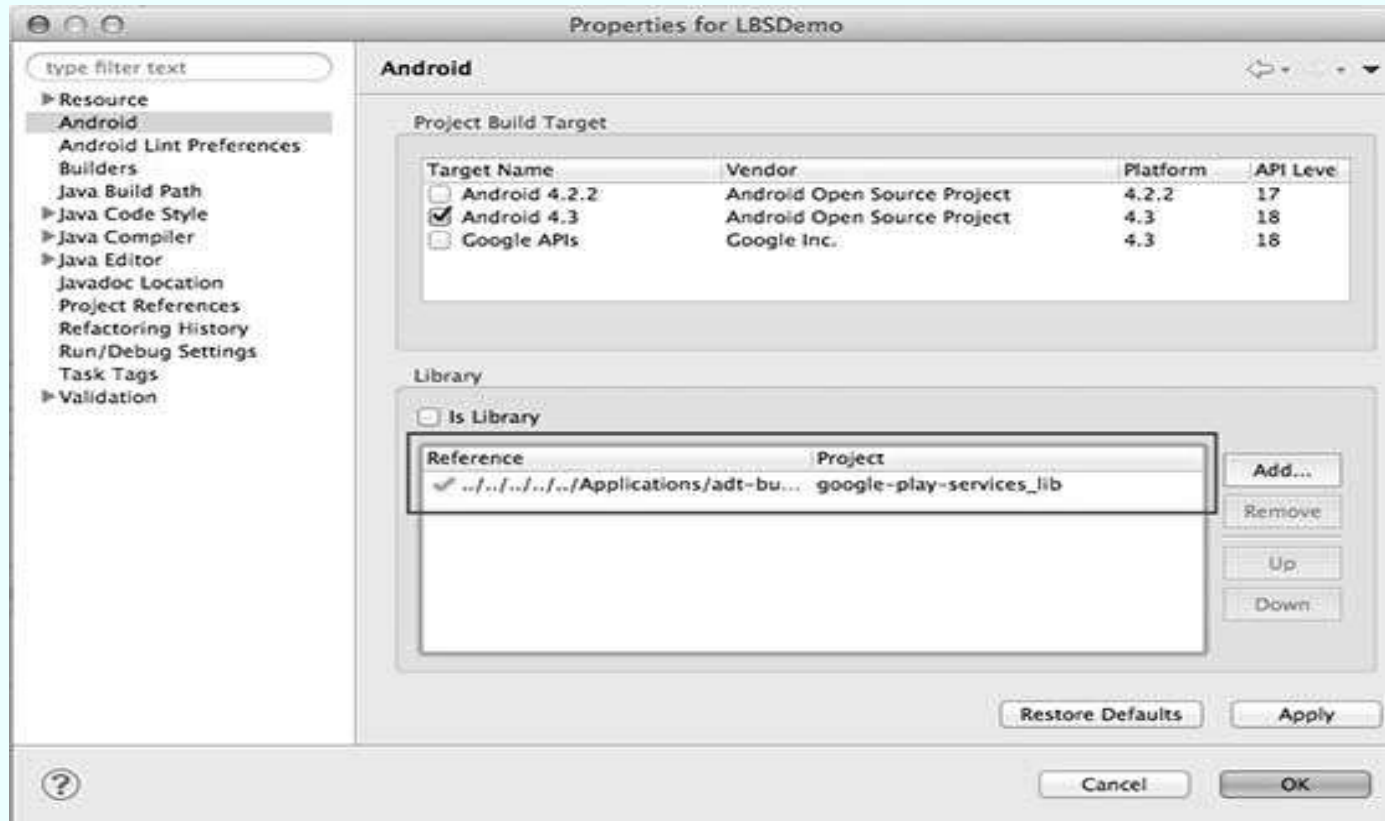
Install Google play service SDK

There are 3 steps:

- In Eclipse, select Window → Android SDK Manager
- Search for **Google Play services** option from the given package list under **Extra** and if it is not installed, then install it.
- Download the google_play_service_lib from its GitHub
- Click **File > Import**, select **Android > Existing Android Code into Workspace**, and browse to the downloaded project to import it.

Add Google play service to project

- Right click on the project and select **Build Path > Configure Build Path > Android >** and then click Add button which will show *google-play-service_lib* option to be added, just double click on it





Example (1)

MainActivity.java (1):

```
package com.example.lbsdemo;
import java.io.IOException;
import java.util.List;
import java.util.Locale;
import com.google.android.gms.common.ConnectionResult;
import com.google.android.gms.common.GooglePlayServicesClient;
import com.google.android.gms.location.LocationClient;
import android.content.Context;
import android.location.Address;
import android.location.Geocoder;
import android.location.Location;
import android.os.AsyncTask;
import android.os.Bundle;
import android.support.v4.app.FragmentActivity;
import android.util.Log;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import android.widget.Toast;

public class MainActivity extends FragmentActivity implements
    GooglePlayServicesClient.ConnectionCallbacks,
    GooglePlayServicesClient.OnConnectionFailedListener{
    LocationClient mLocationClient;
    private TextView addressLabel;
    private TextView locationLabel;
    private Button getLocationBtn;
    private Button disconnectBtn;
    private Button connectBtn;
```



Example (2)

MainActivity.java (2):

```
@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
    locationLabel = (TextView) findViewById(R.id.locationLabel);
    addressLabel = (TextView) findViewById(R.id.addressLabel);
    getLocationBtn = (Button) findViewById(R.id.getLocation);
    getLocationBtn.setOnClickListener(new View.OnClickListener() {
        public void onClick(View view) {
            displayCurrentLocation();
        }
    });
    disconnectBtn = (Button) findViewById(R.id.disconnect);
    disconnectBtn.setOnClickListener(new View.OnClickListener() {
        public void onClick(View view) {
            mLocationClient.disconnect();
            locationLabel.setText("Got disconnected....");
        }
    });
    connectBtn = (Button) findViewById(R.id.connect);
    connectBtn.setOnClickListener(new View.OnClickListener() {
        public void onClick(View view) {
            mLocationClient.connect();
            locationLabel.setText("Got connected....");
        }
    });
    // Create the LocationRequest object
    mLocationClient = new LocationClient(this, this, this);
}
```



Example (3)

MainActivity.java (3):

```
@Override
protected void onStart() {
    super.onStart();
    // Connect the client.
    mLocationClient.connect();
    locationLabel.setText("Got connected....");
}
@Override
protected void onStop() {
    // Disconnect the client.
    mLocationClient.disconnect();
    super.onStop();
    locationLabel.setText("Got disconnected....");
}
@Override
public void onConnected(Bundle dataBundle) {
    // Display the connection status
    Toast.makeText(this, "Connected", Toast.LENGTH_SHORT).show();
}
@Override
public void onDisconnected() {
    // Display the connection status
    Toast.makeText(this, "Disconnected. Please re-connect.",
        Toast.LENGTH_SHORT).show();
}
@Override
public void onConnectionFailed(ConnectionResult connectionResult) {
    // Display the error code on failure
    Toast.makeText(this, "Connection Failure : " +
        connectionResult.getErrorCode(), Toast.LENGTH_SHORT).show();
}
```



Example (4)

MainActivity.java (4):

```
public void displayCurrentLocation() {
    // Get the current location's latitude & longitude
    Location currentLocation = mLocationClient.getLastLocation();
    String msg = "Current Location: " +
        Double.toString(currentLocation.getLatitude()) + "," +
        Double.toString(currentLocation.getLongitude());
    // Display the current location in the UI
    locationLabel.setText(msg);
    // To display the current address in the UI
    (new GetAddressTask(this)).execute(currentLocation);
}

private class GetAddressTask extends AsyncTask<Location, Void,
String>{
    Context mContext;

    public GetAddressTask(Context context) {
        super();
        mContext = context;
    }

    @Override
    protected void onPostExecute(String address) {
        // Display the current address in the UI
        addressLabel.setText(address);
    }
}
```



Example (5)

MainActivity.java (5):

```
@Override
protected String doInBackground(Location... params) {
    Geocoder geocoder = new Geocoder(mContext, Locale.getDefault());
    // Get the current location from the input parameter list
    Location loc = params[0];
    // Create a list to contain the result address
    List<Address> addresses = null;
    try {
        addresses = geocoder.getFromLocation(loc.getLatitude(),
            loc.getLongitude(), 1);
    } catch (IOException e1) {
        Log.e("LocationSampleActivity", "IO Exc in getFromLocation()")
        e1.printStackTrace();
        return ("IO Exception trying to get address");
    } catch (IllegalArgumentException e2) {
        // Error message to post in the log
        String errorString = "Illegal arguments " +
            Double.toString(loc.getLatitude()) + ", " +
            Double.toString(loc.getLongitude()) + " passed to address serv
        Log.e("LocationSampleActivity", errorString);
        e2.printStackTrace();
        return errorString;
    }
}
```



Example (6)

MainActivity.java (6):

```
// If the reverse geocode returned an address
if (addresses != null && addresses.size() > 0) {
    // Get the first address
    Address address = addresses.get(0);
    /*
     * Format the first line of address (if available),
     * city, and country name.
     */
    String addressText = String.format("%s, %s, %s",
        // If there's a street address, add it
        address.getMaxAddressLineIndex() > 0 ?
        address.getAddressLine(0) : "",
        // Locality is usually a city
        address.getLocality(),
        // The country of the address
        address.getCountryName());
    // Return the text
    return addressText;
} else {
    return "No address found";
}
}
} // AsyncTask class
}
```



Example (7)

activity_main.xml:

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="fill_parent"
    android:layout_height="fill_parent"
    android:orientation="vertical" >
```

```
    <Button android:id="@+id/getLocation"
        android:layout_width="fill_parent"
        android:layout_height="wrap_content"
        android:text="@string/get_location"/>
```

```
    <Button android:id="@+id/disconnect"
        android:layout_width="fill_parent"
        android:layout_height="wrap_content"
        android:text="@string/disconnect"/>
```

```
    <Button android:id="@+id/connect"
        android:layout_width="fill_parent"
        android:layout_height="wrap_content"
        android:text="@string/connect"/>
```

```
    <TextView
        android:id="@+id/locationLabel"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>
```

```
    <TextView
        android:id="@+id/addressLabel"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>
```

```
</LinearLayout>
```



Example (8)

strings.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <string name="app_name">LBSDemo</string>
    <string name="action_settings">Settings</string>
    <string name="hello_world">Hello world!</string>
    <string name="get_location">Get Location</string>
    <string name="disconnect">Disconnect Service</string>
    <string name="connect">Connect Service</string>
</resources>
```

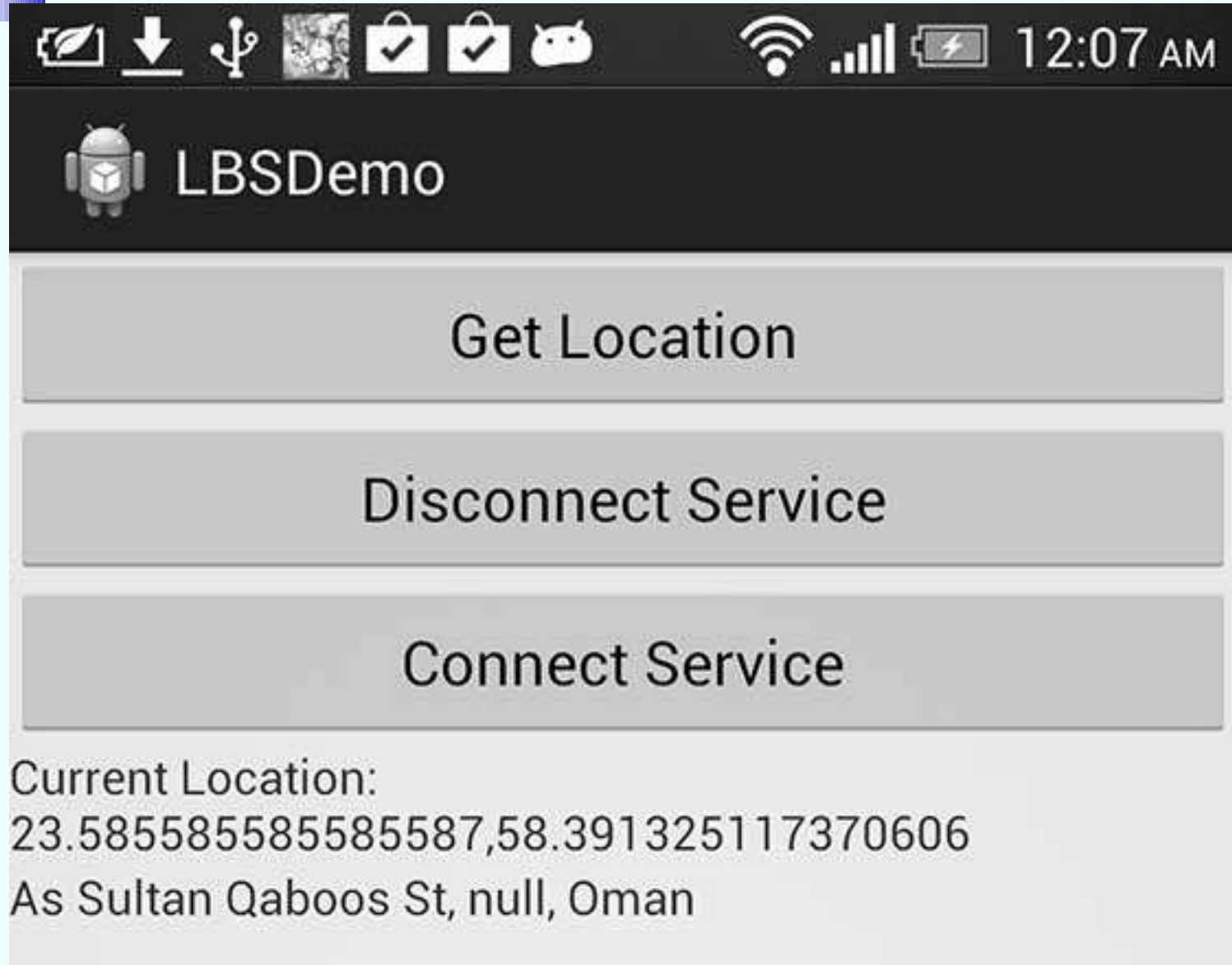


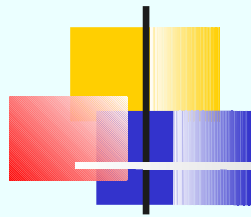
Example (9)

AndroidManifest.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
package="com.example.lbsdemo"
android:versionCode="1"
android:versionName="1.0" >
    <uses-sdk
        android:minSdkVersion="8"
        android:targetSdkVersion="17" />
    <uses-permission
        android:name="android.permission.ACCESS_COARSE_LOCATION"/>
    <application
        android:allowBackup="true"
        android:icon="@drawable/ic_launcher"
        android:label="@string/app_name"
        android:theme="@style/AppTheme" >
        <activity
            android:name="com.example.lbsdemo.MainActivity"
            android:label="@string/app_name" >
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>
</manifest>
```

Example (10)

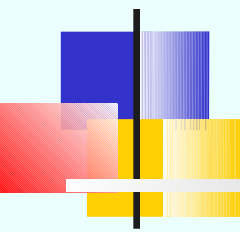




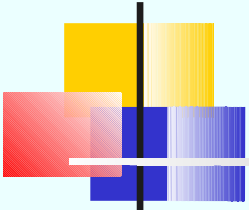
Exercises

App of running tracking:

- Count the time and distance of moving based on the location of an user



Sending email



Intent Object for sending email

Action

```
Intent emailIntent = new Intent(Intent.ACTION_SEND);
```

Data/Type

```
emailIntent.setData(Uri.parse("mailto:"));  
emailIntent.setType("text/plain");
```

Extra

```
emailIntent.putExtra(Intent.EXTRA_EMAIL , new  
    String[]{"recipient@example.com"});  
emailIntent.putExtra(Intent.EXTRA_SUBJECT, "subject of email");  
emailIntent.putExtra(Intent.EXTRA_TEXT , "body of email");
```



Example (1)

MainActivity.java (1):

```
package com.example.sendemaildemo;
import android.net.Uri;
import android.os.Bundle;
import android.app.Activity;
import android.content.Intent;
import android.util.Log;
import android.view.Menu;
import android.view.View;
import android.widget.Button;
import android.widget.Toast;

public class MainActivity extends Activity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        Button startBtn = (Button) findViewById(R.id.sendEmail);
        startBtn.setOnClickListener(new View.OnClickListener() {
            public void onClick(View view) {
                sendEmail();
            }
        });
    }
}
```



Example (2)

MainActivity.java (2):

```
protected void sendEmail() {
    Log.i("Send email", "");
    String[] TO = {"amrood.admin@gmail.com"};
    String[] CC = {"mcmohd@gmail.com"};
    Intent emailIntent = new Intent(Intent.ACTION_SEND);
    emailIntent.setData(Uri.parse("mailto:"));
    emailIntent.setType("text/plain");
    emailIntent.putExtra(Intent.EXTRA_EMAIL, TO);
    emailIntent.putExtra(Intent.EXTRA_CC, CC);
    emailIntent.putExtra(Intent.EXTRA_SUBJECT, "Your subject");
    emailIntent.putExtra(Intent.EXTRA_TEXT, "Email message goes here")
    try {
        startActivity(Intent.createChooser(emailIntent, "Send mail..."));
        finish();
        Log.i("Finished sending email...", "");
    } catch (android.content.ActivityNotFoundException ex) {
        Toast.makeText(MainActivity.this, "There is no email
            client installed.", Toast.LENGTH_SHORT).show();
    }
}

@Override
public boolean onCreateOptionsMenu(Menu menu) {
    //Inflate the menu; this adds items to the action bar if it is pr
    getMenuInflater().inflate(R.menu.main, menu);
    return true;
}
}
```



Example (3)

activity_main.xml:

```
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="fill_parent"
android:layout_height="fill_parent"
android:orientation="vertical" >

    <Button android:id="@+id/sendEmail"
        android:layout_width="fill_parent"
        android:layout_height="wrap_content"
        android:text="@string/compose_email"/>
</LinearLayout>
```



Example (4)

strings.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <string name="app_name">SendEmailDemo</string>
    <string name="hello_world">Hello world!</string>
    <string name="action_settings">Settings</string>
    <string name="compose_email">Compose Email</string>
</resources>
```



Example (5)

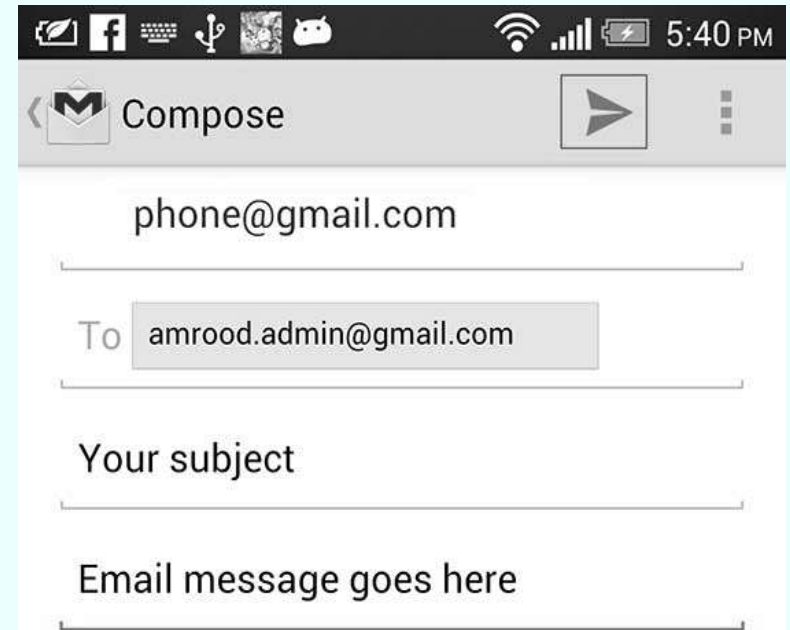
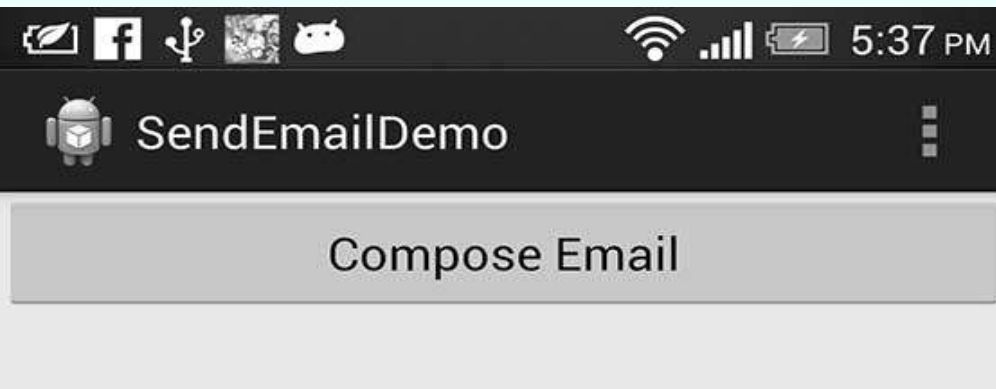
AndroidManifest.xml:

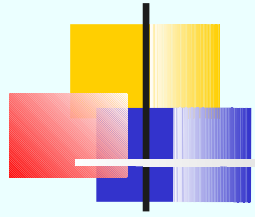
```
<application
  android:allowBackup="true"
  android:icon="@drawable/ic_launcher"
  android:label="@string/app_name"
  android:theme="@style/AppTheme" >

  <activity
    android:name="com.example.sendemaildemo.MainActivity"
    android:label="@string/app_name" >

    <intent-filter>
      <action android:name="android.intent.action.MAIN" />
      <category android:name="android.intent.category.LAUNCHER" />
    </intent-filter>
  </activity>
</application>
```

Example (6)

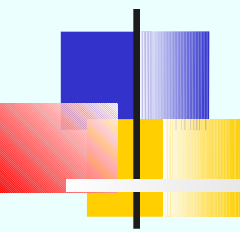




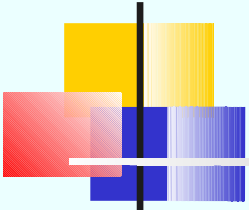
Exercises

Create an app of client mail:

- Receive and display mail inbox
- Sending email
- Notification if there is new mail arrived



Sending SMS



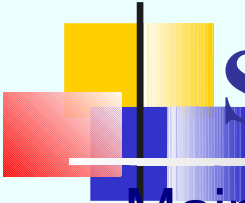
Using SMSManager

Create object

```
SmsManager smsManager = SmsManager.getDefault();
```

Sending method

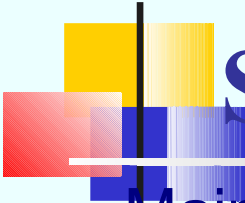
```
smsManager.sendTextMessage("phoneNo", null, "SMS text", null, null);
```



SMSManager example (1)

MainActivity.java (1):

```
package com.example.sendsmsdemo;
import android.os.Bundle;
import android.app.Activity;
import android.telephony.SmsManager;
import android.util.Log;
import android.view.Menu;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
public class MainActivity extends Activity {
    Button sendBtn;
    EditText txtphoneNo;
    EditText txtMessage;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        sendBtn = (Button) findViewById(R.id.btnSendSMS);
        txtphoneNo = (EditText) findViewById(R.id.editTextPhoneNo);
        txtMessage = (EditText) findViewById(R.id.editTextSMS);
        sendBtn.setOnClickListener(new View.OnClickListener() {
            public void onClick(View view) {
                sendSMSMessage();
            }
        });
    }
}
```



SMSManager example (2)

MainActivity.java (2):

```
protected void sendSMSMessage() {
    Log.i("Send SMS", "");
    String phoneNo = txtphoneNo.getText().toString();
    String message = txtMessage.getText().toString();
    try {
        SmsManager smsManager = SmsManager.getDefault();
        smsManager.sendTextMessage(phoneNo, null, message, null, null);
        Toast.makeText(getApplicationContext(), "SMS sent.",
            Toast.LENGTH_LONG).show();
    } catch (Exception e) {
        Toast.makeText(getApplicationContext(),
            "SMS failed, please try again.", Toast.LENGTH_LONG).show();
        e.printStackTrace();
    }
}

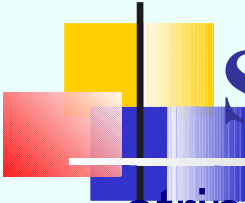
@Override
public boolean onCreateOptionsMenu(Menu menu) {
    //Inflate the menu; it adds items to action bar if it is present
    getMenuInflater().inflate(R.menu.main, menu);
    return true;
}
```



SMSManager example (3)

activity_main.xml:

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="fill_parent"
    android:layout_height="fill_parent"
    android:orientation="vertical" >
    <TextView
        android:id="@+id/textViewPhoneNo"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="@string/phone_label" />
    <EditText
        android:id="@+id/editTextPhoneNo"
        android:layout_width="fill_parent"
        android:layout_height="wrap_content"
        android:inputType="phone"/>
    <TextView
        android:id="@+id/textViewMessage"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="@string/sms_label" />
    <EditText
        android:id="@+id/editTextSMS"
        android:layout_width="fill_parent"
        android:layout_height="wrap_content"
        android:inputType="textMultiLine"/>
    <Button android:id="@+id/btnSendSMS"
        android:layout_width="fill_parent"
        android:layout_height="wrap_content"
        android:text="@string/send_sms_label"/>
</LinearLayout>
```



SMSManager example (4)

strings.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <string name="app_name">SendSMSDemo</string>
    <string name="action_settings">Settings</string>
    <string name="hello_world">Hello world!</string>
    <string name="phone_label">Enter Phone Number:</string>
    <string name="sms_label">Enter SMS Message:</string>
    <string name="send_sms_label">Send SMS</string>
</resources>
```



SMSManager example (5)

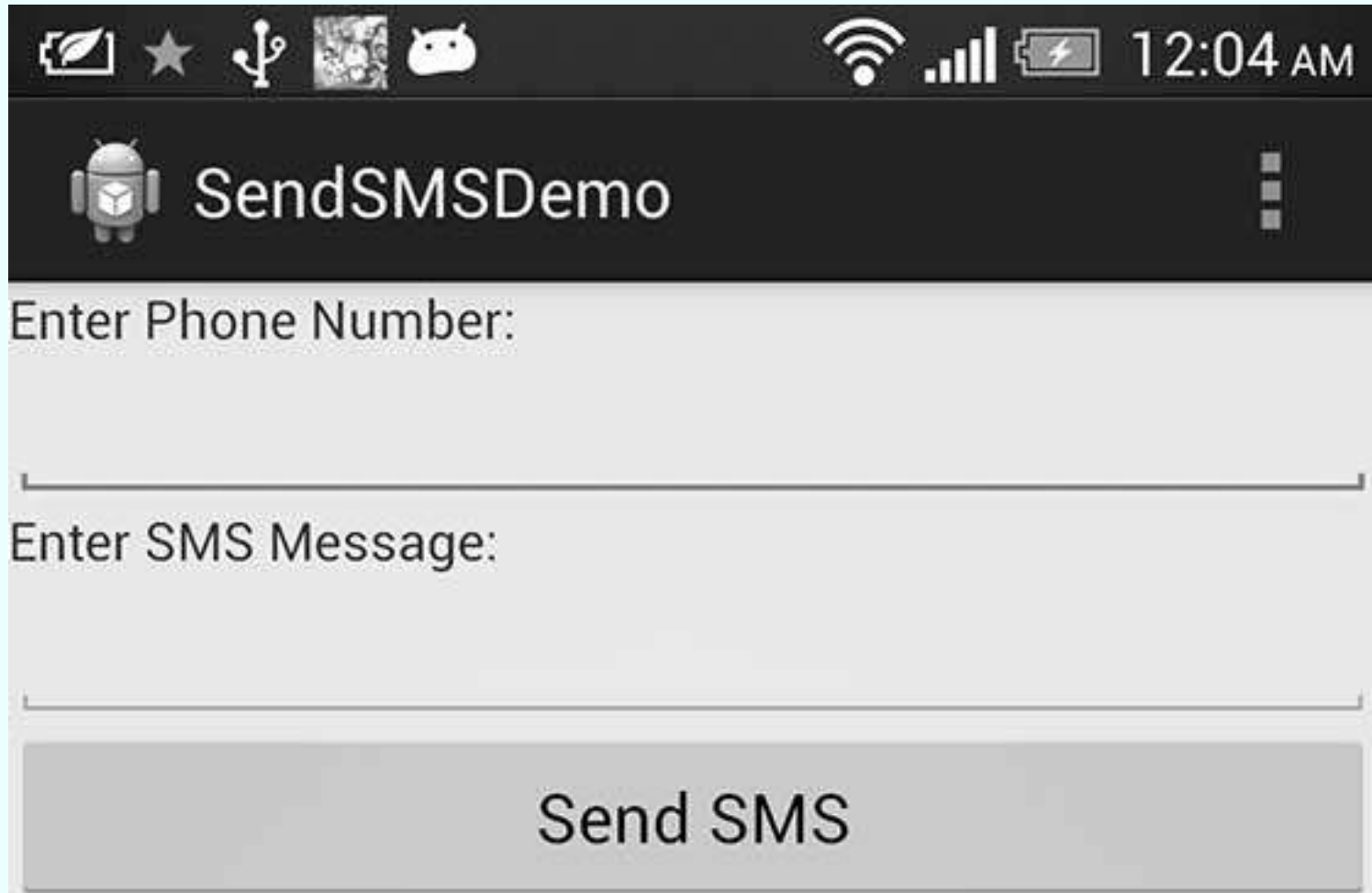
AndroidManifest.xml:

```
<uses-sdk
    android:minSdkVersion="8"
    android:targetSdkVersion="17" />

<uses-permission android:name="android.permission.SEND_SMS" />

<application
    android:allowBackup="true"
    android:icon="@drawable/ic_launcher"
    android:label="@string/app_name"
    android:theme="@style/AppTheme" >
    <activity
        android:name="com.example.sendsmsdemo.MainActivity"
        android:label="@string/app_name" >
        <intent-filter>
            <action android:name="android.intent.action.MAIN" />
            <category android:name="android.intent.category.LAUNCHER"/>
        </intent-filter>
    </activity>
</application>
```

SMSManager example (6)

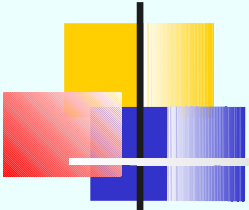


The screenshot shows an Android application window titled "SendSMSDemo". The status bar at the top displays various icons including a leaf, a star, a USB symbol, a group of people, an Android robot, Wi-Fi, cellular signal, battery, and the time "12:04 AM". The application interface has a dark header bar with the Android robot icon and the title "SendSMSDemo". Below the header, there are two text input fields. The first field is labeled "Enter Phone Number:" and the second field is labeled "Enter SMS Message:". At the bottom of the interface is a large, light gray button labeled "Send SMS".

Enter Phone Number:

Enter SMS Message:

Send SMS



Using Intent object

Action

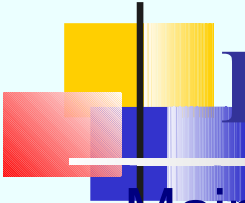
```
Intent smsIntent = new Intent(Intent.ACTION_VIEW);
```

Data/Type

```
smsIntent.setData(Uri.parse("smsto:"));  
smsIntent.setType("vnd.android-dir/mms-sms");
```

Extra

```
smsIntent.putExtra("address" , new String("0123456789;3393993300"));  
smsIntent.putExtra("sms_body" , "Test SMS to Angilla");
```



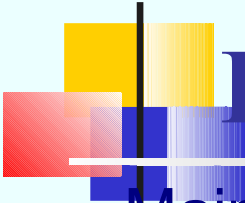
Intent example (1)

MainActivity.java (1):

```
package com.example.sendsmsdemo;
import android.net.Uri;
import android.os.Bundle;
import android.app.Activity;
import android.content.Intent;
import android.util.Log;
import android.view.Menu;
import android.view.View;
import android.widget.Button;
import android.widget.Toast;

public class MainActivity extends Activity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        Button startBtn = (Button) findViewById(R.id.sendSMS);
        startBtn.setOnClickListener(new View.OnClickListener() {
            public void onClick(View view) {
                sendSMS();
            }
        });
    }
}
```



Intent example (2)

MainActivity.java (2):

```
protected void sendSMS() {
    Log.i("Send SMS", "");
    Intent smsIntent = new Intent(Intent.ACTION_VIEW);
    smsIntent.setData(Uri.parse("smsto:"));
    smsIntent.setType("vnd.android-dir/mms-sms");
    smsIntent.putExtra("address" , new String ("0123456789"));
    smsIntent.putExtra("sms_body" , "Test SMS to Angilla");
    try {
        startActivity(smsIntent);
        finish();
        Log.i("Finished sending SMS...", "");
    } catch (android.content.ActivityNotFoundException ex) {
        Toast.makeText(MainActivity.this,
            "SMS failed, please try again later.", Toast.LENGTH_SHORT).show();
    }
}

@Override
public boolean onCreateOptionsMenu(Menu menu) {
    // Inflate the menu; it adds items to the action bar if it is present
    getMenuInflater().inflate(R.menu.main, menu);
    return true;
}
}
```

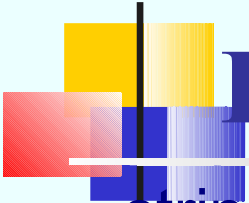


Intent example (3)

activity_main.xml:

```
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="fill_parent"
android:layout_height="fill_parent"
android:orientation="vertical" >

    <Button android:id="@+id/sendSMS"
        android:layout_width="fill_parent"
        android:layout_height="wrap_content"
        android:text="@string/compose_sms"/>
</LinearLayout>
```



Intent example (4)

strings.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <string name="app_name">SendSMSDemo</string>
    <string name="hello_world">Hello world!</string>
    <string name="action_settings">Settings</string>
    <string name="compose_sms">Compose SMS</string>
</resources>
```



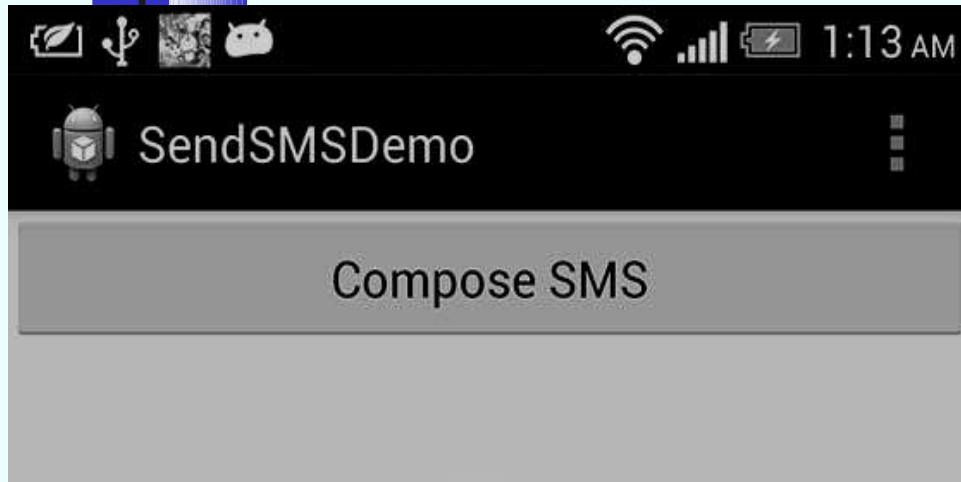
Intent example (5)

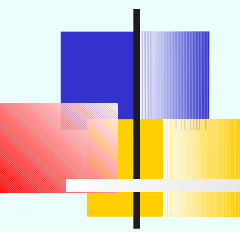
AndroidManifest.xml:

```
<uses-sdk
    android:minSdkVersion="8"
    android:targetSdkVersion="17" />

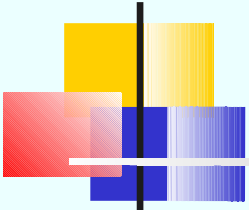
<application
    android:allowBackup="true"
    android:icon="@drawable/ic_launcher"
    android:label="@string/app_name"
    android:theme="@style/AppTheme" >
    <activity
        android:name="com.example.sendsmsdemo.MainActivity"
        android:label="@string/app_name" >
        <intent-filter>
            <action android:name="android.intent.action.MAIN" />
            <category android:name="android.intent.category.LAUNCHER" />
        </intent-filter>
    </activity>
</application>
```

Intent example (6)





Phone calls



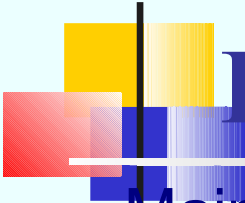
Intent object

Action

```
Intent phoneIntent = new Intent(Intent.ACTION_CALL);
```

Data/Type

```
phoneIntent.setData(Uri.parse("tel:91-800-001-0101"));
```



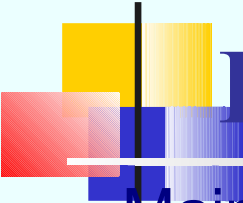
Example (1)

MainActivity.java (1):

```
package com.example.phonecalldemo;
import android.net.Uri;
import android.os.Bundle;
import android.app.Activity;
import android.content.Intent;
import android.util.Log;
import android.view.Menu;
import android.view.View;
import android.widget.Button;
import android.widget.Toast;

public class MainActivity extends Activity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        Button startBtn = (Button) findViewById(R.id.makeCall);
        startBtn.setOnClickListener(new View.OnClickListener() {
            public void onClick(View view) {
                makeCall();
            }
        });
    }
}
```



Example (2)

MainActivity.java (2):

```
protected void makeCall() {
    Log.i("Make call", "");
    Intent phoneIntent = new Intent(Intent.ACTION_CALL);
    phoneIntent.setData(Uri.parse("tel:91-800-001-0101"));
    try {
        startActivity(phoneIntent);
        finish();
        Log.i("Finished making a call...", "");
    } catch (android.content.ActivityNotFoundException ex) {
        Toast.makeText(MainActivity.this, "Call failed, please try again
            later.", Toast.LENGTH_SHORT).show();
    }
}

@Override
public boolean onCreateOptionsMenu(Menu menu) {
    // Inflate the menu; this adds items to the action bar if it is present
    getMenuInflater().inflate(R.menu.main, menu);
    return true;
}
}
```

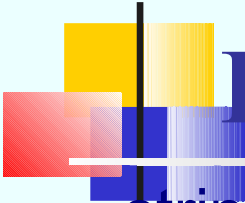


Example (3)

activity_main.xml:

```
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="fill_parent"
android:layout_height="fill_parent"
android:orientation="vertical" >

    <Button android:id="@+id/makeCall"
        android:layout_width="fill_parent"
        android:layout_height="wrap_content"
        android:text="@string/make_call"/>
</LinearLayout>
```



Example (4)

strings.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <string name="app_name">PhoneCallDemo</string>
    <string name="hello_world">Hello world!</string>
    <string name="action_settings">Settings</string>
    <string name="make_call">Call 91-800-001-0101</string>
</resources>
```



Example (5)

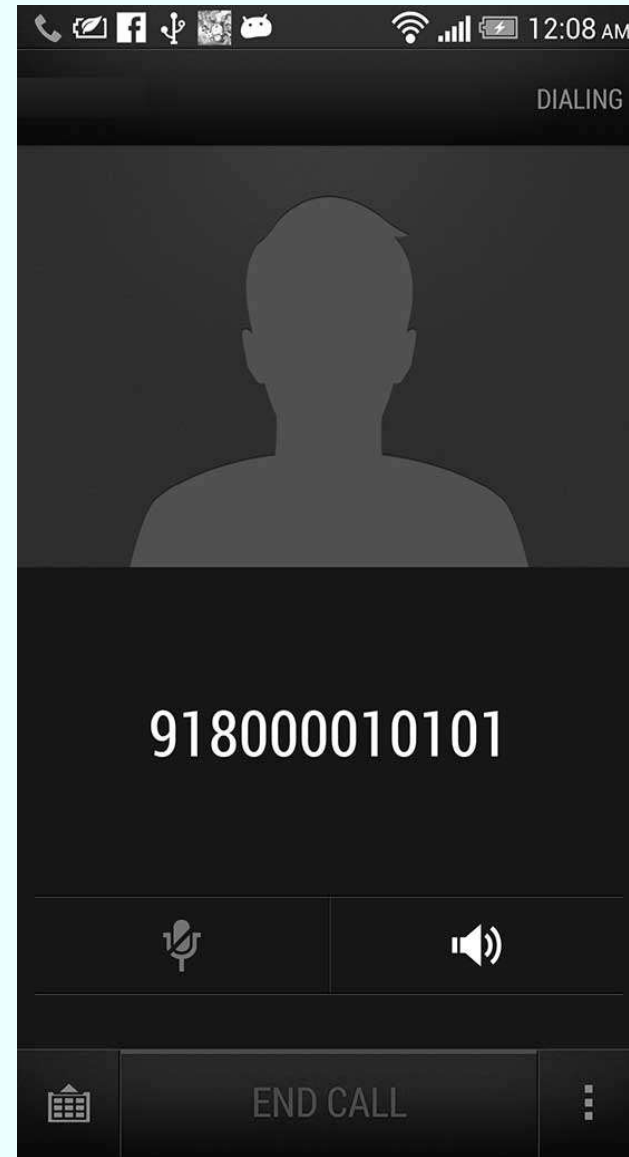
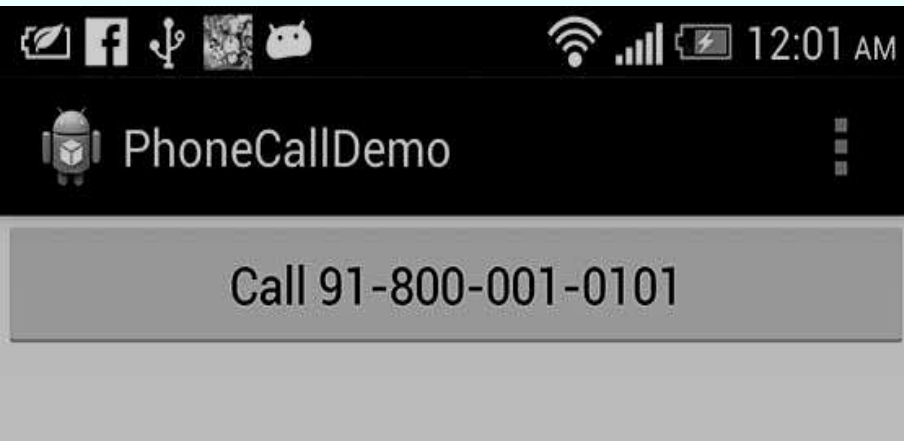
AndroidManifest.xml:

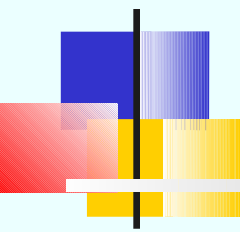
```
<uses-sdk
    android:minSdkVersion="8"
    android:targetSdkVersion="17" />

<uses-permission android:name="android.permission.CALL_PHONE" />
<uses-permission android:name="android.permission.READ_PHONE_STATE"/>

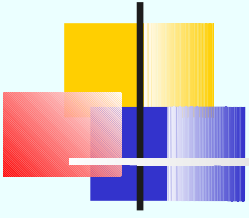
<application
    android:allowBackup="true"
    android:icon="@drawable/ic_launcher"
    android:label="@string/app_name"
    android:theme="@style/AppTheme" >
    <activity
        android:name="com.example.phonecalldemo.MainActivity"
        android:label="@string/app_name" >
        <intent-filter>
            <action android:name="android.intent.action.MAIN" />
            <category android:name="android.intent.category.LAUNCHER" />
        </intent-filter>
    </activity>
</application>
```

Example (6)





Alert dialog



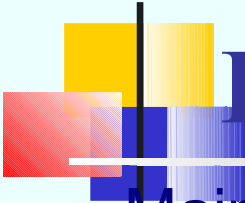
AlertDialog object

Create

```
AlertDialog.Builder alertDialogBuilder = new AlertDialog.Builder(this)
```

Using

```
alertDialogBuilder.setPositiveButton(CharSequence text,  
    DialogInterface.OnClickListener listener)  
alertDialogBuilder.setNegativeButton(CharSequence text,  
    DialogInterface.OnClickListener listener)
```



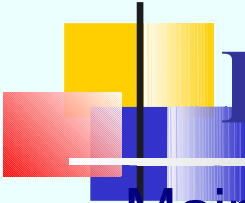
Example (1)

MainActivity.java (1):

```
package com.example.alertdialog;
import com.example.alertdialog.*;
import android.os.Bundle;
import android.app.Activity;
import android.app.AlertDialog;
import android.content.DialogInterface;
import android.content.Intent;
import android.view.Menu;
import android.view.View;

public class MainActivity extends Activity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        // Inflate the menu; this adds items to the action bar
        // if it is present.
        getMenuInflater().inflate(R.menu.main, menu);
        return true;
    }
}
```



Example (2)

MainActivity.java (2):

```
public void open(View view){
    AlertDialog.Builder alertDialogBuilder = new AlertDialog.Builder(
        alertDialogBuilder.setMessage(R.string.decision);
        alertDialogBuilder.setPositiveButton(R.string.positive_button,
            new DialogInterface.OnClickListener() {
                @Override
                public void onClick(DialogInterface arg0, int arg1) {
                    Intent positiveActivity = new
                        Intent(getApplicationContext(),
                            com.example.alertdialog.PositiveActivity.class);
                    startActivity(positiveActivity);
                }
            });
        alertDialogBuilder.setNegativeButton(R.string.negative_button,
            new DialogInterface.OnClickListener() {
                @Override
                public void onClick(DialogInterface dialog, int which) {
                    Intent negativeActivity = new
                        Intent(getApplicationContext(),
                            com.example.alertdialog.NegativeActivity.class);
                    startActivity(negativeActivity);
                }
            });
        AlertDialog alertDialog = alertDialogBuilder.create();
        alertDialog.show();
    }
```



Example (3)

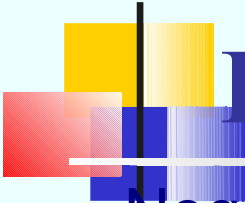
PositiveActivity.java:

```
package com.example.alertdialog;
import android.os.Bundle;
import android.app.Activity;
import android.view.Menu;

public class PositiveActivity extends Activity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_positive);
    }

    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        // Inflate the menu; this adds items to the action bar
        // if it is present.
        getMenuInflater().inflate(R.menu.positive, menu);
        return true;
    }
}
```



Example (4)

NegativeActivity.java:

```
package com.example.alertdialog;
import android.os.Bundle;
import android.app.Activity;
import android.view.Menu;

public class NegativeActivity extends Activity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_negative);
    }

    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        // Inflate the menu; this adds items to the action bar
        // if it is present.
        getMenuInflater().inflate(R.menu.negative, menu);
        return true;
    }
}
```



Example (5)

activity_main.xml:

```
<RelativeLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:paddingBottom="@dimen/activity_vertical_margin"
android:paddingLeft="@dimen/activity_horizontal_margin"
android:paddingRight="@dimen/activity_horizontal_margin"
android:paddingTop="@dimen/activity_vertical_margin"
tools:context=".MainActivity" >
```

```
    <Button
        android:id="@+id/button1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignParentTop="true"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="170dp"
        android:onClick="open"
        android:text="@string/hello_world" />
```

```
</RelativeLayout>
```



Example (6)

activity_positive.xml:

```
<RelativeLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:paddingBottom="@dimen/activity_vertical_margin"
android:paddingLeft="@dimen/activity_horizontal_margin"
android:paddingRight="@dimen/activity_horizontal_margin"
android:paddingTop="@dimen/activity_vertical_margin"
tools:context=".PositiveActivity" >

    <TextView
        android:id="@+id/textView1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignParentLeft="true"
        android:layout_alignParentTop="true"
        android:layout_marginLeft="14dp"
        android:layout_marginTop="20dp"
        android:text="@string/positive"
        android:textAppearance="?android:attr/textAppearanceLarge" />
</RelativeLayout>
```

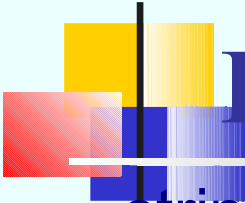


Example (7)

activity_negative.xml:

```
<RelativeLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:paddingBottom="@dimen/activity_vertical_margin"
android:paddingLeft="@dimen/activity_horizontal_margin"
android:paddingRight="@dimen/activity_horizontal_margin"
android:paddingTop="@dimen/activity_vertical_margin"
tools:context=".NegativeActivity" >

    <TextView
        android:id="@+id/textView1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignParentLeft="true"
        android:layout_alignParentTop="true"
        android:layout_marginLeft="14dp"
        android:layout_marginTop="17dp"
        android:text="@string/negative"
        android:textAppearance="?android:attr/textAppearanceLarge" />
</RelativeLayout>
```



Example (8)

strings.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
<string name="app_name">AlertDialog</string>
<string name="action_settings">Settings</string>
<string name="hello_world">Hello world!</string>
<string name="title_activity_positive">PositiveActivity</string>
<string name="title_activity_negative">NegativeActivity</string>
<string name="positive">Positive Activity</string>
<string name="negative">Negative Activity</string>
<string name="decision">Are you sure, you wanted to make this
decision</string>
<string name="positive_button">+ive</string>
<string name="negative_button">-ive</string>
</resources>
```

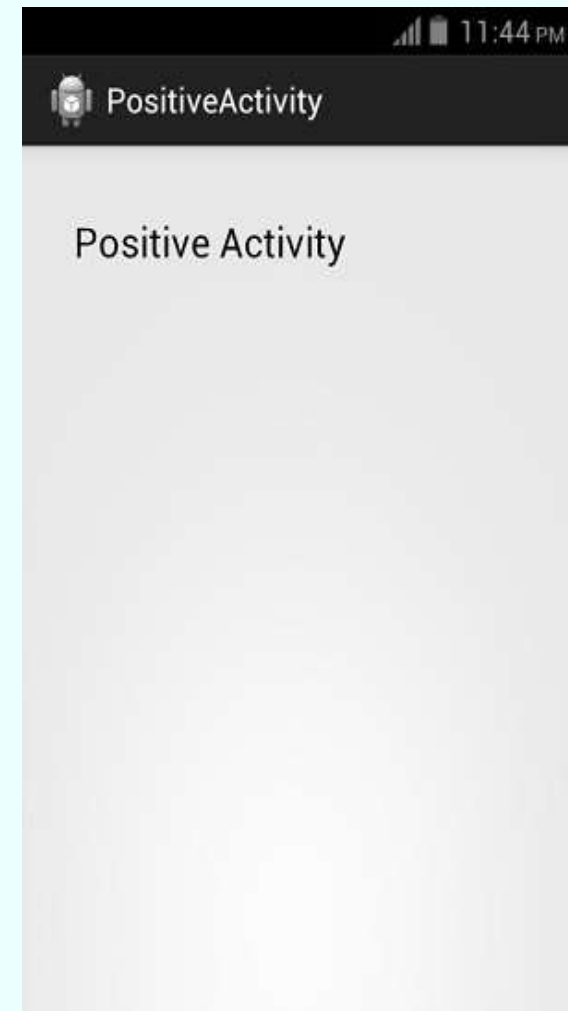
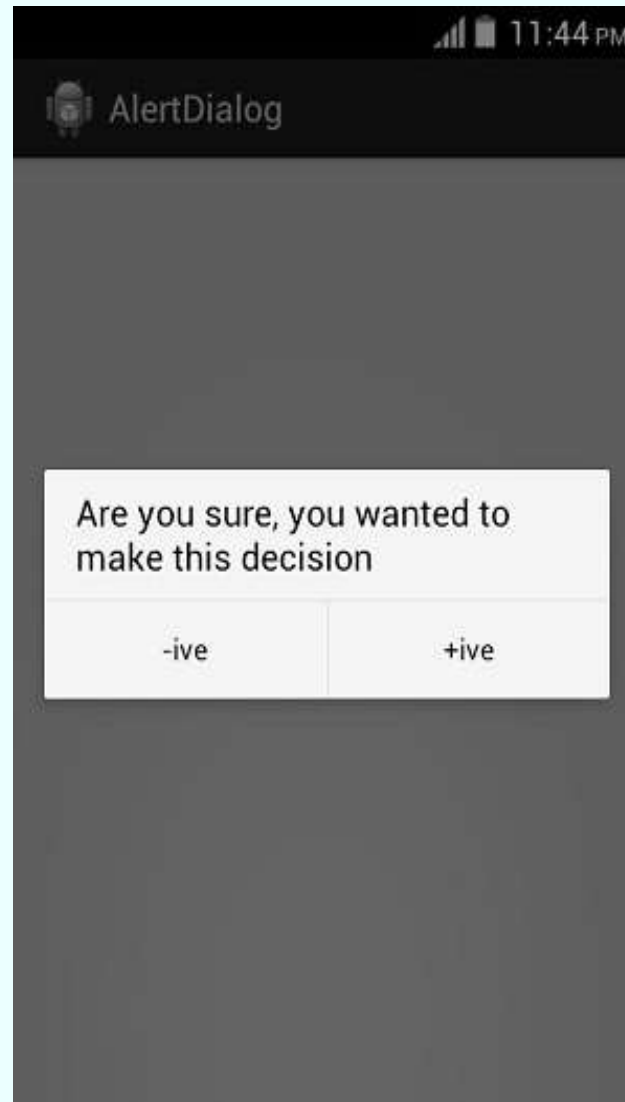
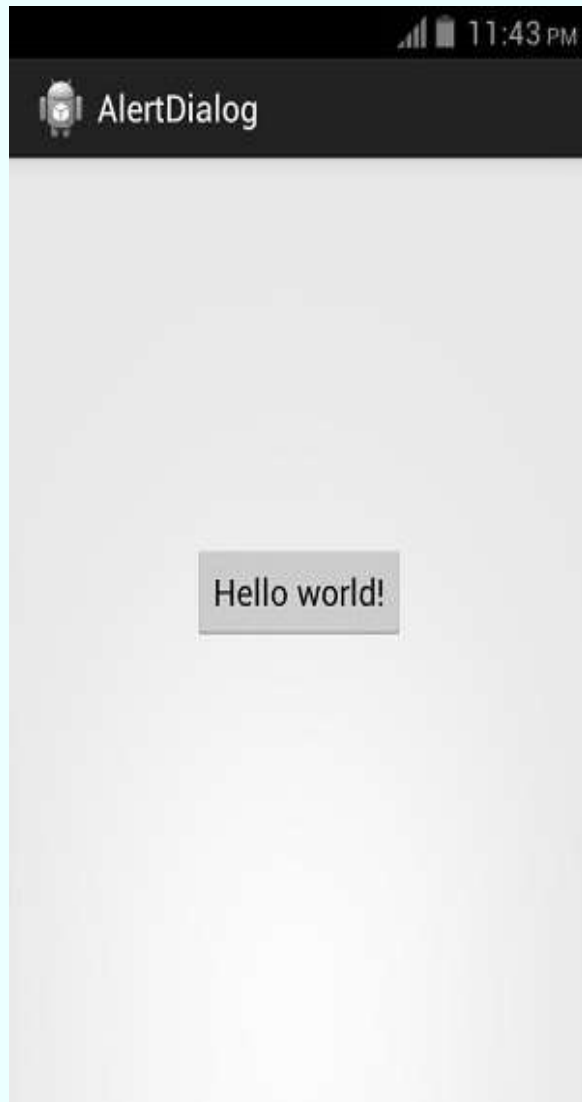


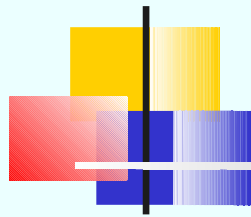
Example (9)

AndroidManifest.xml:

```
<application
  android:allowBackup="true"
  android:icon="@drawable/ic_launcher"
  android:label="@string/app_name"
  android:theme="@style/AppTheme" >
  <activity
    android:name="com.example.alertdialog.MainActivity"
    android:label="@string/app_name" >
    <intent-filter>
      <action android:name="android.intent.action.MAIN" />
      <category android:name="android.intent.category.LAUNCHER" />
    </intent-filter>
  </activity>
  <activity
    android:name="com.example.alertdialog.PositiveActivity"
    android:label="@string/title_activity_positive" >
  </activity>
  <activity
    android:name="com.example.alertdialog.NegativeActivity"
    android:label="@string/title_activity_negative" >
  </activity>
</application>
```

Example (10)

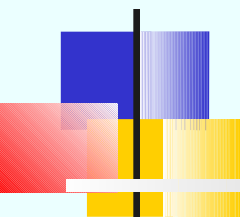




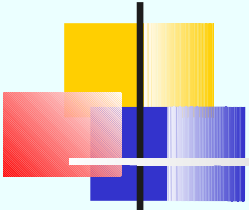
Exercises

Within the example of student management:

- When delete a student, create a dialog to ask if user be sure to delete it.



Animations



Tween animation

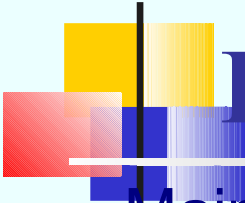
Create

```
Animation animation = AnimationUtils.loadAnimation(  
    getApplicationContext(), R.anim.myanimation);
```

Using

```
ImageView image1 = (ImageView) findViewById(R.id.imageView1);  
image.startAnimation(animation);
```

Note: need to create an animation xml file in
res/anim



Example (1)

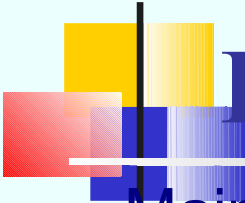
MainActivity.java (1):

```
package com.example.animation;
import com.example.animation.R;
import android.os.Bundle;
import android.app.Activity;
import android.view.Menu;
import android.view.MenuItem;
import android.view.animation.Animation;
import android.view.animation.AnimationUtils;
import android.widget.ImageView;

public class MainActivity extends Activity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }

    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        // Inflate the menu; this adds items to the action bar
        // if it is present.
        getMenuInflater().inflate(R.menu.main, menu);
        return true;
    }
}
```



Example (2)

MainActivity.java (2):

```
public boolean onOptionsItemSelected(MenuItem item) {
    super.onOptionsItemSelected(item);
    switch(item.getItemId()) {
        case R.id.zoomInOut:
            ImageView image = (ImageView)findViewById(R.id.imageView1);
            Animation animation =
                AnimationUtils.loadAnimation(getApplicationContext(),
                    R.anim.myanimation);
            image.startAnimation(animation);
            return true;
        case R.id.rotate360:
            ImageView image1 = (ImageView)findViewById(R.id.imageView1);
            Animation animation1 =
                AnimationUtils.loadAnimation(getApplicationContext(),
                    R.anim.clockwise);
            image1.startAnimation(animation1);
            return true;
        case R.id.fadeInOut:
            ImageView image2 = (ImageView)findViewById(R.id.imageView1);
            Animation animation2 =
                AnimationUtils.loadAnimation(getApplicationContext(),
                    R.anim.fade);
            image2.startAnimation(animation2);
            return true;
    }
    return false;
}
```

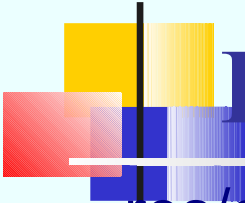


Example (3)

activity_main.xml:

```
<RelativeLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:gravity="top"
android:paddingBottom="@dimen/activity_vertical_margin"
android:paddingLeft="@dimen/activity_horizontal_margin"
android:paddingRight="@dimen/activity_horizontal_margin"
android:paddingTop="@dimen/activity_vertical_margin"
tools:context=".MainActivity" >
```

```
<ImageView
android:id="@+id/imageView1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_alignParentTop="true"
android:layout_centerHorizontal="true"
android:layout_marginTop="179dp"
android:src="@drawable/ic_launcher" />
</RelativeLayout>
```

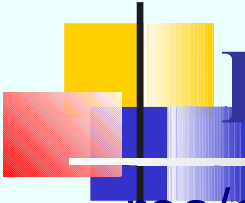


Example (4)

res/anim/myanimation.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<set xmlns:android="http://schemas.android.com/apk/res/android">
    <scale xmlns:android="http://schemas.android.com/apk/res/android"
        android:fromXScale="0.5"
        android:toXScale="3.0"
        android:fromYScale="0.5"
        android:toYScale="3.0"
        android:duration="5000"
        android:pivotX="50%"
        android:pivotY="50%" >
    </scale>

    <scale xmlns:android="http://schemas.android.com/apk/res/android"
        android:startOffset="5000"
        android:fromXScale="3.0"
        android:toXScale="0.5"
        android:fromYScale="3.0"
        android:toYScale="0.5"
        android:duration="5000"
        android:pivotX="50%"
        android:pivotY="50%" >
    </scale>
</set>
```

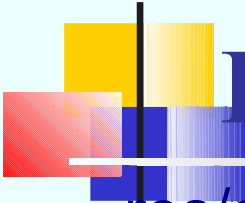


Example (5)

res/anim/clockwise.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<set xmlns:android="http://schemas.android.com/apk/res/android">
<rotate xmlns:android="http://schemas.android.com/apk/res/android"
android:fromDegrees="0"
android:toDegrees="360"
android:pivotX="50%"
android:pivotY="50%"
android:duration="5000" >
</rotate>

<rotate xmlns:android="http://schemas.android.com/apk/res/android"
android:startOffset="5000"
android:fromDegrees="360"
android:toDegrees="0"
android:pivotX="50%"
android:pivotY="50%"
android:duration="5000" >
</rotate>
</set>
```



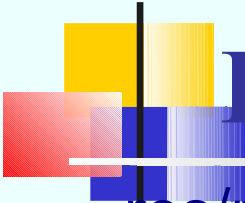
Example (6)

res/anim/fade.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<set xmlns:android="http://schemas.android.com/apk/res/android"
    android:interpolator="@android:anim/accelerate_interpolator" >

    <alpha
        android:fromAlpha="0"
        android:toAlpha="1"
        android:duration="2000" >

    <alpha
        android:startOffset="2000"
        android:fromAlpha="1"
        android:toAlpha="0"
        android:duration="2000" >
    </alpha>
</set>
```



Example (7)

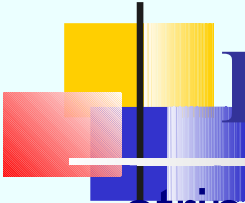
res/menu/main.xml:

```
<menu xmlns:android="http://schemas.android.com/apk/res/android" >

    <item
        android:id="@+id/rotate360"
        android:orderInCategory="100"
        android:showAsAction="never"
        android:title="@string/rotate_String"/>

    <item
        android:id="@+id/zoomInOut"
        android:orderInCategory="100"
        android:title="@string/zoom_In_Out"/>

    <item
        android:id="@+id/fadeInOut"
        android:orderInCategory="100"
        android:title="@string/fade_String"/>
</menu>
```



Example (8)

strings.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
<string name="app_name">Animation</string>
<string name="action_settings">Settings</string>
<string name="hello_world">Hello world!</string>
<string name="zoom_In_Out">Zoom In/Out</string>
<string name="rotate_String">Clockwise/Anti Clockwise</string>
<string name="fade_String">Fade In/Out</string>
</resources>
```



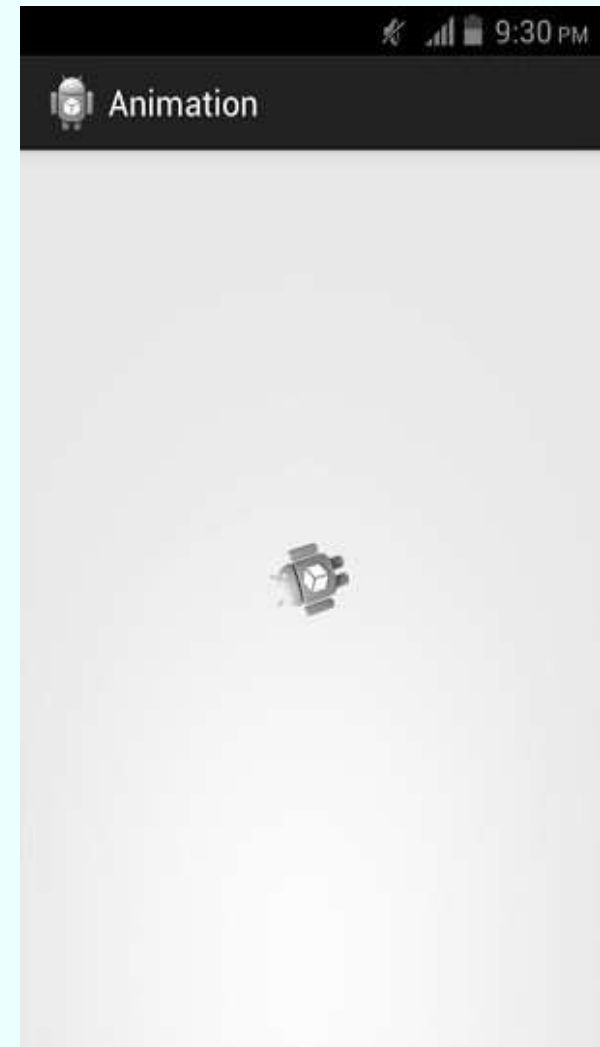
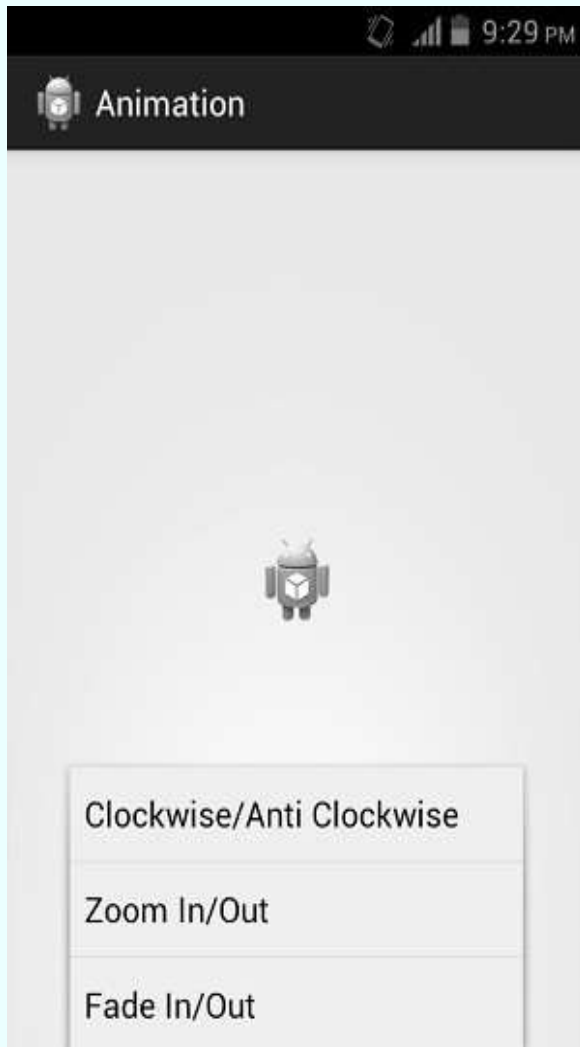
Example (9)

AndroidManifest.xml:

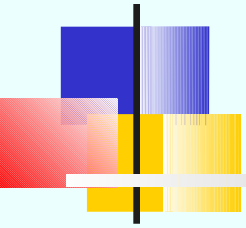
```
<application
  android:allowBackup="true"
  android:icon="@drawable/ic_launcher"
  android:label="@string/app_name"
  android:theme="@style/AppTheme" >

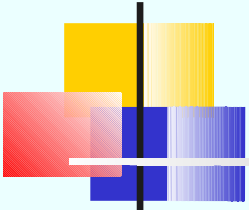
  <activity
    android:name="com.example.animation.MainActivity"
    android:label="@string/app_name" >
    <intent-filter>
      <action android:name="android.intent.action.MAIN" />
      <category android:name="android.intent.category.LAUNCHER" />
    </intent-filter>
  </activity>
</application>
```

Example (10)



Audio





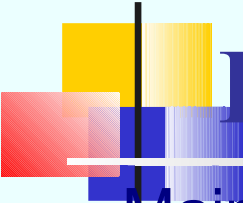
Audio capture

Create

```
MediaRecorder myAudioRecorder = new MediaRecorder();
```

Using

```
myAudioRecorder.setAudioSource(MediaRecorder.AudioSource.MIC);  
myAudioRecorder.setOutputFormat(MediaRecorder.OutputFormat.THREE_GPP);  
myAudioRecorder.setAudioEncoder(MediaRecorder.OutputFormat.AMR_NB);  
myAudioRecorder.setOutputFile(outputFile);  
myAudioRecorder.prepare();  
myAudioRecorder.start();
```

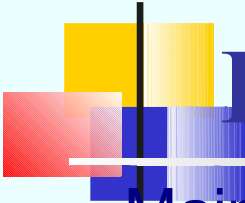


Example (1)

MainActivity.java (1):

```
public class MainActivity extends Activity {
    private MediaRecorder myAudioRecorder;
    private String outputFile = null;
    private Button start, stop, play;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        start = (Button)findViewById(R.id.button1);
        stop = (Button)findViewById(R.id.button2);
        play = (Button)findViewById(R.id.button3);
        stop.setEnabled(false);
        play.setEnabled(false);
        outputFile = Environment.getExternalStorageDirectory().
            getAbsolutePath() + "/myrecording.3gp";
        myAudioRecorder = new MediaRecorder();
        myAudioRecorder.setAudioSource(MediaRecorder.AudioSource.MIC);
        myAudioRecorder.setOutputFormat(
            MediaRecorder.OutputFormat.THREE_GPP);
        myAudioRecorder.setAudioEncoder(
            MediaRecorder.OutputFormat.AMR_NB);
        myAudioRecorder.setOutputFile(outputFile);
    }
}
```



Example (2)

MainActivity.java (2):

```
public void start(View view) {
    try {
        myAudioRecorder.prepare();
        myAudioRecorder.start();
    } catch (Exception e) {
        e.printStackTrace();
    }
    start.setEnabled(false);
    stop.setEnabled(true);
    Toast.makeText(getApplicationContext(), "Recording started",
        Toast.LENGTH_LONG).show();
}

public void stop(View view) {
    myAudioRecorder.stop();
    myAudioRecorder.release();
    myAudioRecorder = null;
    stop.setEnabled(false);
    play.setEnabled(true);
    Toast.makeText(getApplicationContext(),
        "Audio recorded successfully", Toast.LENGTH_LONG).show();
}

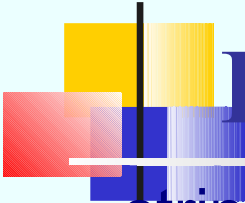
public void play(View view) throws IllegalArgumentException,
    SecurityException, IllegalStateException, IOException {
    MediaPlayer m = new MediaPlayer();
    m.setDataSource(outputFile);
    m.prepare();
    m.start();
    Toast.makeText(getApplicationContext(), "Playing audio",
        Toast.LENGTH_LONG).show();
}
```



Example (3)

activity_main.xml:

```
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent" android:layout_height="match_parent"
    <TextView
        android:id="@+id/textView1"
        android:layout_width="wrap_content" android:layout_height="wrap_content"
        android:layout_alignParentLeft="true" android:layout_alignParentRight="true"
        android:layout_alignParentTop="true" android:layout_marginTop="32dp"
        android:text="@string/Recording"
        android:textAppearance="?android:attr/textAppearanceMedium" />
    <ImageView
        android:id="@+id/imageView1" android:layout_width="100dp"
        android:layout_height="100dp" android:layout_below="@+id/textView1"
        android:layout_centerHorizontal="true" android:layout_marginTop="37dp"
        android:scaleType="fitXY"
        android:src="@android:drawable/presence_audio_online" />
    <Button
        android:id="@+id/button1" android:layout_width="wrap_content"
        android:layout_height="wrap_content" android:layout_below="@+id/imageView1"
        android:layout_marginTop="67dp" android:layout_toLeftOf="@+id/imageView1"
        android:onClick="start" android:text="@string/start" />
    <Button
        android:id="@+id/button2" android:layout_width="wrap_content"
        android:layout_height="wrap_content" android:layout_alignBaseline="@+id/button1"
        android:layout_alignBottom="@+id/button1" android:layout_alignRight="@+id/textView1"
        android:layout_marginRight="40dp" android:onClick="stop"
        android:text="@string/stop" />
    <Button
        android:id="@+id/button3" style="?android:attr/buttonStyleSmall"
        android:layout_width="wrap_content" android:layout_height="wrap_content"
        android:layout_below="@+id/button2" android:layout_centerHorizontal="true" />
```



Example (4)

strings.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
<string name="app_name">AudioCapture</string>
<string name="action_settings">Settings</string>
<string name="hello_world">Hello world!</string>
<string name="Recording">Android Audio Recording Application</string>
<string name="start">start</string>
<string name="stop">stop</string>
<string name="play">play</string>
</resources>
```



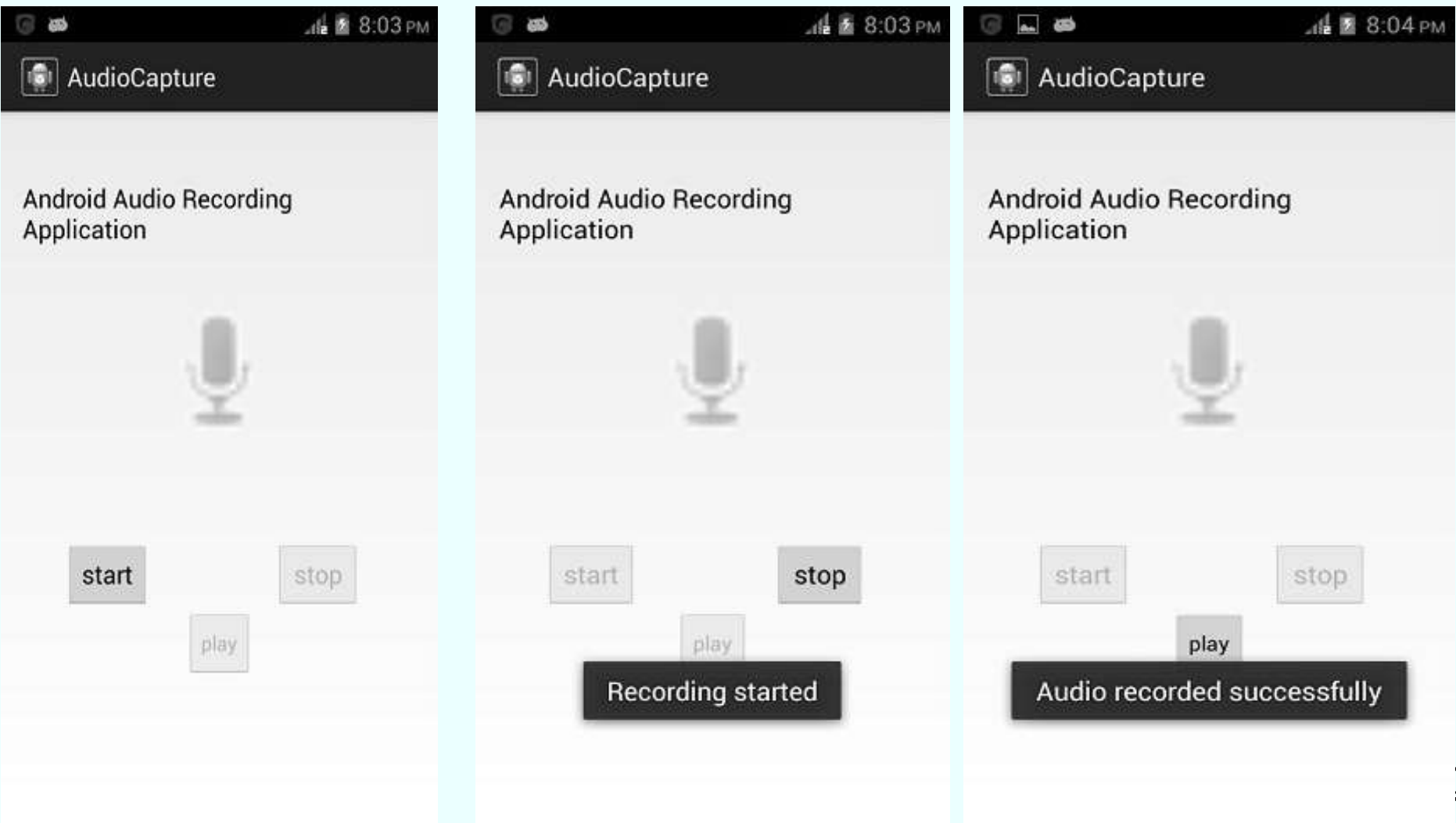
Example (5)

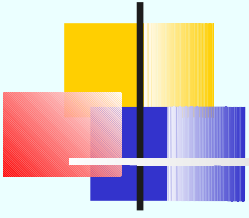
AndroidManifest.xml:

```
<uses-permission
android:name="android.permission.WRITE_EXTERNAL_STORAGE"/>
<uses-permission android:name="android.permission.RECORD_AUDIO" />

<application
android:allowBackup="true"
android:icon="@drawable/ic_launcher"
android:label="@string/app_name"
android:theme="@style/AppTheme" >
    <activity
        android:name="com.example.audiocapture.MainActivity"
        android:label="@string/app_name" >
        <intent-filter>
            <action android:name="android.intent.action.MAIN" />
            <category android:name="android.intent.category.LAUNCHER" />
        </intent-filter>
    </activity>
</application>
```

Example (6)





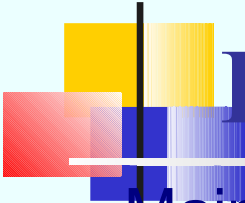
Audio manager

Create

```
AudioManager myAudioManager =  
    (AudioManager) getSystemService (Context.AUDIO_SERVICE) ;
```

Using

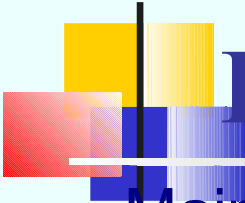
```
myAudioManager.setRingerMode (AudioManager.RINGER_MODE_VIBRATE) ;
```



Example (1)

MainActivity.java (1):

```
public class MainActivity extends Activity {  
    private Button Vibrate, Ring, Silent, Mode;  
    private TextView Status;  
    private AudioManager myAudioManager;  
  
    @Override  
    protected void onCreate(Bundle savedInstanceState) {  
        super.onCreate(savedInstanceState);  
        setContentView(R.layout.activity_main);  
        Vibrate = (Button)findViewById(R.id.button2);  
        Ring = (Button)findViewById(R.id.button4);  
        Silent = (Button)findViewById(R.id.button3);  
        Mode = (Button)findViewById(R.id.button1);  
        Status = (TextView)findViewById(R.id.textView2);  
        myAudioManager =  
            (AudioManager) getSystemService(Context.AUDIO_SERVICE);  
    }  
}
```



Example (2)

MainActivity.java (2):

```
public void vibrate(View view){
    myAudioManager.setRingerMode(AudioManager.RINGER_MODE_VIBRATE);
}
public void ring(View view){
    myAudioManager.setRingerMode(AudioManager.RINGER_MODE_NORMAL);
}
public void silent(View view){
    myAudioManager.setRingerMode(AudioManager.RINGER_MODE_SILENT);
}
public void mode(View view){
    int mod = myAudioManager.getRingerMode();
    if(mod == AudioManager.RINGER_MODE_NORMAL){
        Status.setText("Current Status: Ring");
    }else if(mod == AudioManager.RINGER_MODE_SILENT){
        Status.setText("Current Status: Silent");
    }else if(mod == AudioManager.RINGER_MODE_VIBRATE){
        Status.setText("Current Status: Vibrate");
    }
}
```



Example (3)

activity_main.xml (1):

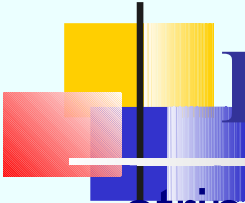
```
<RelativeLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:paddingBottom="@dimen/activity_vertical_margin"
android:paddingLeft="@dimen/activity_horizontal_margin"
android:paddingRight="@dimen/activity_horizontal_margin"
android:paddingTop="@dimen/activity_vertical_margin"
tools:context=".MainActivity" >
<TextView
android:id="@+id/textView1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_alignParentTop="true"
android:layout_centerHorizontal="true"
android:layout_marginTop="25dp"
android:text="@string/audio"
android:textAppearance="?android:attr/textAppearanceLarge" />
<Button
android:id="@+id/button2"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_alignBaseline="@+id/button3"
android:layout_alignBottom="@+id/button3"
android:layout_alignRight="@+id/textView1"
android:onClick="vibrate"
android:text="@string/Vibrate" />
```



Example (4)

activity_main.xml (2):

```
<Button
android:id="@+id/button3"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_alignParentBottom="true"
android:layout_alignParentLeft="true"
android:layout_marginBottom="144dp"
android:layout_marginLeft="40dp"
android:layout_toLeftOf="@+id/button2"
android:onClick="silent"
android:text="@string/Silent" />
<Button
android:id="@+id/button4"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_alignBaseline="@+id/button1"
android:layout_alignBottom="@+id/button1"
android:layout_toRightOf="@+id/button1"
android:onClick="ring"
android:text="@string/Ring" />
<Button
android:id="@+id/button1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_above="@+id/button2"
android:layout_alignLeft="@+id/button3"
android:layout_marginBottom="15dp"
android:onClick="mode"
android:text="@string/Mode" />
<TextView
android:id="@+id/textView2"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_below="@+id/textView1"
android:layout_centerHorizontal="true"
android:layout_marginTop="46dp"
android:text="@string/Status"
android:textAppearance="?android:attr/textAppearanceMedium" />
</RelativeLayout>
```



Example (5)

strings.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
<string name="app_name">AudioManager</string>
<string name="action_settings">Settings</string>
<string name="hello_world">Hello world!</string>
<string name="audio">Set Audio Profiles</string>
<string name="Ring">Ring</string>
<string name="Vibrate">Vibrate</string>
<string name="Silent">Silent</string>
<string name="Mode">Current Mode</string>
<string name="Status">Current Status</string>
</resources>
```

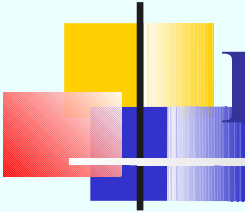


Example (6)

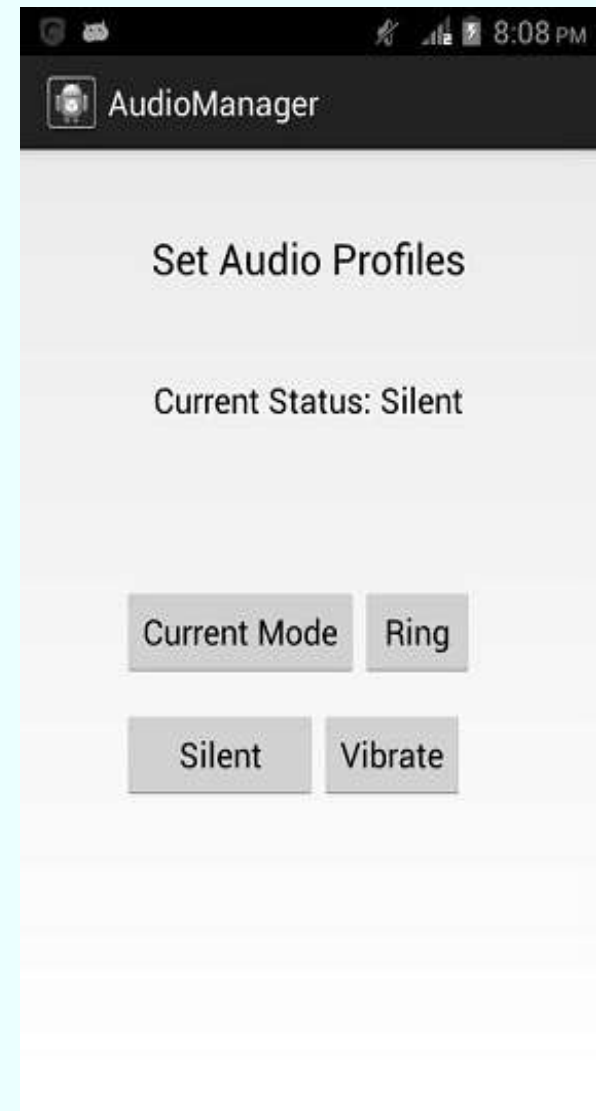
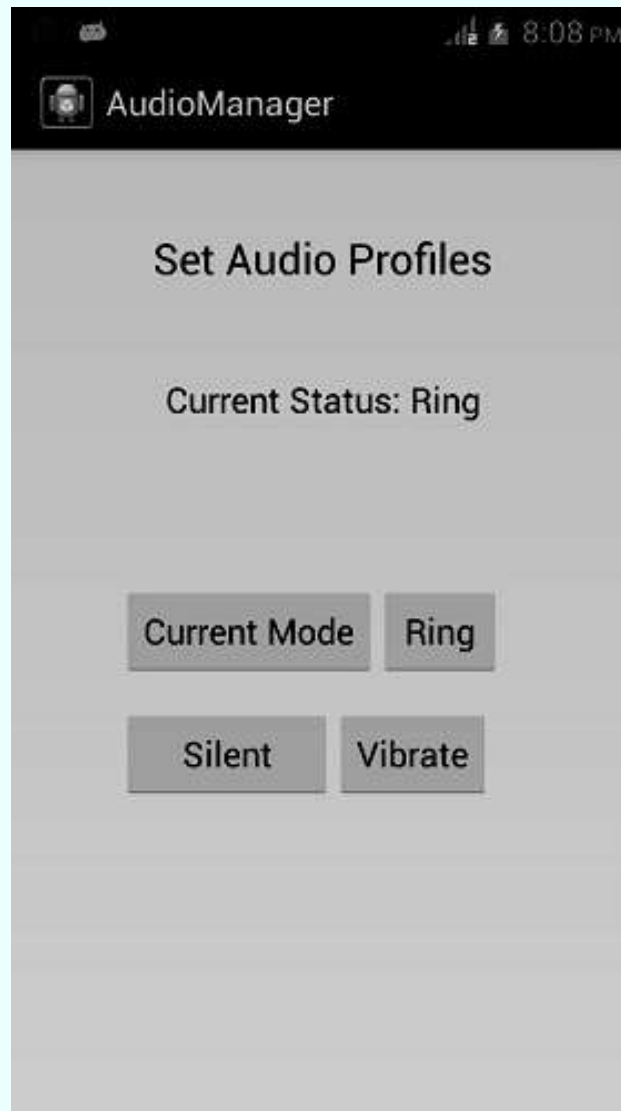
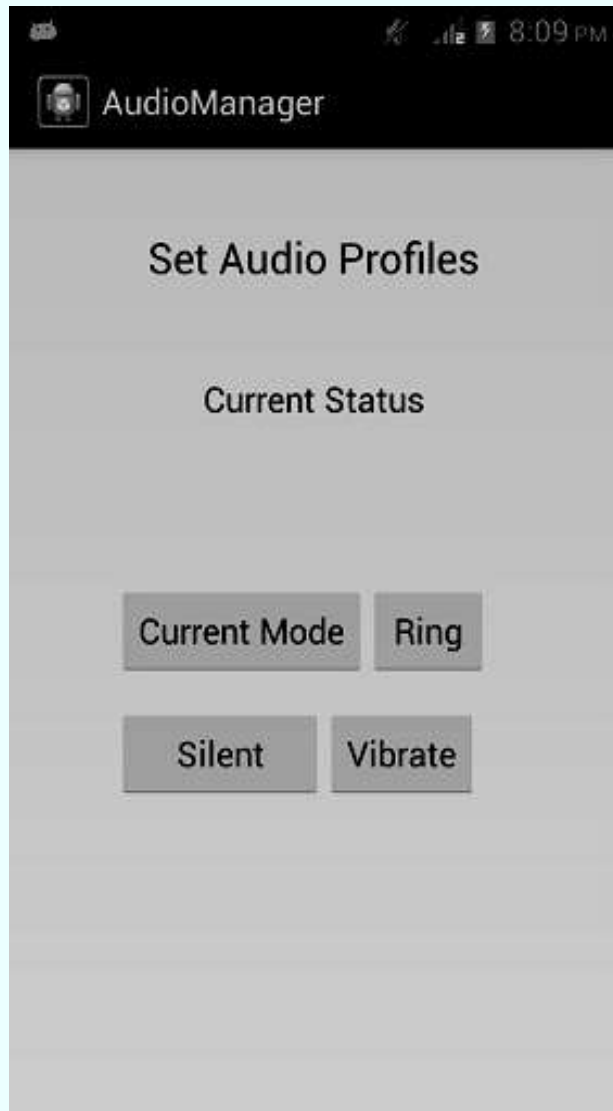
AndroidManifest.xml:

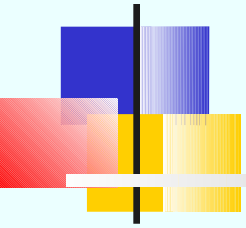
```
<application
  android:allowBackup="true"
  android:icon="@drawable/ic_launcher"
  android:label="@string/app_name"
  android:theme="@style/AppTheme" >

  <activity
    android:name="com.example.audiomanager.MainActivity"
    android:label="@string/app_name" >
    <intent-filter>
      <action android:name="android.intent.action.MAIN" />
      <category android:name="android.intent.category.LAUNCHER" />
    </intent-filter>
  </activity>
</application>
```

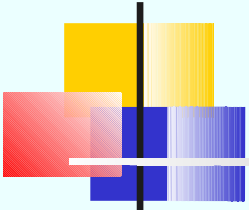


Example (7)





Bluetooth



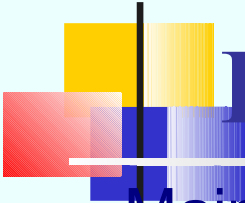
BluetoothAdapter

Create

```
BluetoothAdapter BA = BluetoothAdapter.getDefaultAdapter();
```

Using

```
Intent turnOn = new Intent(BluetoothAdapter.ACTION_REQUEST_ENABLE);  
startActivityForResult(turnOn, 0);
```



Example (1)

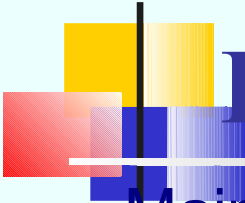
MainActivity.java (1):

```
public class MainActivity extends Activity {
    private Button On, Off, Visible, list;
    private BluetoothAdapter BA;
    private Set<BluetoothDevice> pairedDevices;
    private ListView lv;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        On = (Button)findViewById(R.id.button1);
        Off = (Button)findViewById(R.id.button2);
        Visible = (Button)findViewById(R.id.button3);
        list = (Button)findViewById(R.id.button4);
        lv = (ListView)findViewById(R.id.listView1);
        BA = BluetoothAdapter.getDefaultAdapter();
    }

    public void off(View view){
        BA.disable();
        Toast.makeText(getApplicationContext(), "Turned off" ,
            Toast.LENGTH_LONG).show();
    }

    public void visible(View view){
        Intent getVisible = new Intent(BluetoothAdapter.
            ACTION_REQUEST_DISCOVERABLE);
        startActivityForResult(getVisible, 0);
    }
}
```



Example (2)

MainActivity.java (2):

```
public void on(View view){
    if (!BA.isEnabled()) {
        Intent turnOn = new
            Intent(BluetoothAdapter.ACTION_REQUEST_ENABLE);
        startActivityForResult(turnOn, 0);
        Toast.makeText(getApplicationContext(), "Turned on"
            , Toast.LENGTH_LONG).show();
    }else{
        Toast.makeText(getApplicationContext(), "Already on",
            Toast.LENGTH_LONG).show();
    }
}

public void list(View view){
    pairedDevices = BA.getBondedDevices();
    ArrayList list = new ArrayList();
    for(BluetoothDevice bt : pairedDevices)
        list.add(bt.getName());
    Toast.makeText(getApplicationContext(), "Showing Paired Devices",
        Toast.LENGTH_SHORT).show();
    final ArrayAdapter adapter = new ArrayAdapter
        (this, android.R.layout.simple_list_item_1, list);
    lv.setAdapter(adapter);
}
```



Example (3)

activity_main.xml (1):

```
<RelativeLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:paddingBottom="@dimen/activity_vertical_margin"
android:paddingLeft="@dimen/activity_horizontal_margin"
android:paddingRight="@dimen/activity_horizontal_margin"
android:paddingTop="@dimen/activity_vertical_margin"
tools:context=".MainActivity" >
```

```
<ScrollView
android:id="@+id/scrollView1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_alignParentBottom="true"
android:layout_alignParentLeft="true"
android:layout_alignParentRight="true"
android:layout_alignParentTop="true" >
```

```
<LinearLayout
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical" >
```

```
<TextView
android:id="@+id/textView1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="@string/app_name"
android:textAppearance="?android:attr/textAppearanceLarge" />
```



Example (4)

activity_main.xml (2):

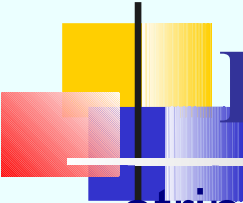
```
<Button
    android:id="@+id/button1"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:onClick="on"
    android:text="@string/on" />
```

```
<Button
    android:id="@+id/button2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:onClick="visible"
    android:text="@string/Visible" />
```

```
<Button
    android:id="@+id/button3"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:onClick="list"
    android:text="@string/List" />
```

```
<Button
    android:id="@+id/button4"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:onClick="off"
    android:text="@string/off" />
```

```
<ListView
    android:id="@+id/listView1"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:visibility="visible" />
</ListView>
</LinearLayout>
</ScrollView>
```



Example (5)

strings.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
<string name="app_name">Bluetooth</string>
<string name="action_settings">Settings</string>
<string name="hello_world">Hello world!</string>
<string name="on">Turn On</string>
<string name="off">Turn Off</string>
<string name="Visible">Get Visible</string>
<string name="List">List Devices</string>
</resources>
```



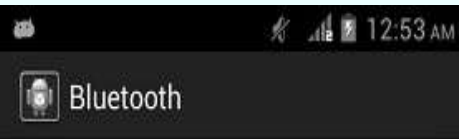
Example (6)

AndroidManifest.xml:

```
<uses-permission android:name="android.permission.BLUETOOTH"/>
<uses-permission android:name="android.permission.BLUETOOTH_ADMIN"/>

<application
    android:allowBackup="true"
    android:icon="@drawable/ic_launcher"
    android:label="@string/app_name"
    android:theme="@style/AppTheme" >
    <activity
        android:name="com.example.bluetooth.MainActivity"
        android:label="@string/app_name" >
        <intent-filter>
            <action android:name="android.intent.action.MAIN" />
            <category android:name="android.intent.category.LAUNCHER" />
        </intent-filter>
    </activity>
</application>
```

Example (7)



Bluetooth

Turn On

Get Visible

List Devices

Turn Off



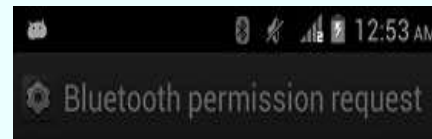
Bluetooth permission request

An app wants to turn on Bluetooth. Allow?

No

Yes

Turned on

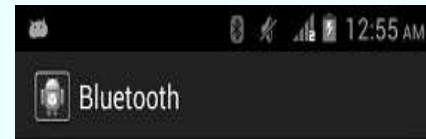


Bluetooth permission request

An app on your phone wants to make your phone discoverable by other Bluetooth devices for 120 seconds. Allow?

No

Yes



Bluetooth

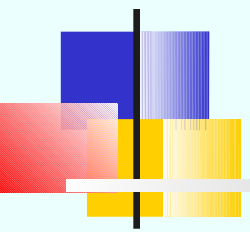
Turn On

Get Visible

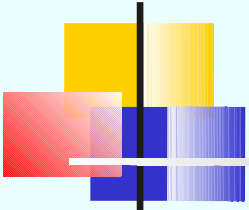
List Devices

Turn Off

Nokia C2-01



Camera



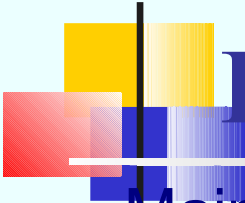
Intent object

Create

```
Intent intent = new  
    Intent(android.provider.MediaStore.ACTION_IMAGE_CAPTURE);
```

Using

```
startActivityForResult(intent, 0)
```



Example (1)

MainActivity.java:

```
public class MainActivity extends Activity {
    ImageView imgFavorite;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        imgFavorite = (ImageView) findViewById(R.id.imageView1);
        imgFavorite.setOnClickListener(new OnClickListener() {
            @Override
            public void onClick(View v) {
                open();
            }
        });
    }
    public void open(){
        Intent intent = new
        Intent(android.provider.MediaStore.ACTION_IMAGE_CAPTURE);
        startActivityForResult(intent, 0);
    }
    @Override
    protected void onActivityResult(int requestCode, int resultCode,
        Intent data) {
        // TODO Auto-generated method stub
        super.onActivityResult(requestCode, resultCode, data);
        Bitmap bp = (Bitmap) data.getExtras().get("data");
        imgFavorite.setImageBitmap(bp);
    }
}
```



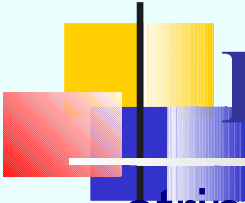
Example (2)

activity_main.xml:

```
<RelativeLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:paddingBottom="@dimen/activity_vertical_margin"
android:paddingLeft="@dimen/activity_horizontal_margin"
android:paddingRight="@dimen/activity_horizontal_margin"
android:paddingTop="@dimen/activity_vertical_margin"
tools:context=".MainActivity">

<ImageView
android:id="@+id/imageView1"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:layout_marginLeft="34dp"
android:layout_marginTop="36dp"
android:contentDescription="@string/hello_world"
android:src="@drawable/ic_launcher" />

<TextView
android:id="@+id/textView1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_alignParentTop="true"
android:layout_alignRight="@+id/imageView1"
android:text="@string/tap"
android:textAppearance="?android:attr/textAppearanceLarge" />
</RelativeLayout>
```



Example (3)

strings.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
<string name="app_name">Camera</string>
<string name="action_settings">Settings</string>
<string name="hello_world">Hello world!</string>
<string name="tap">Tap the image to open the camera!!</string>
</resources>
```



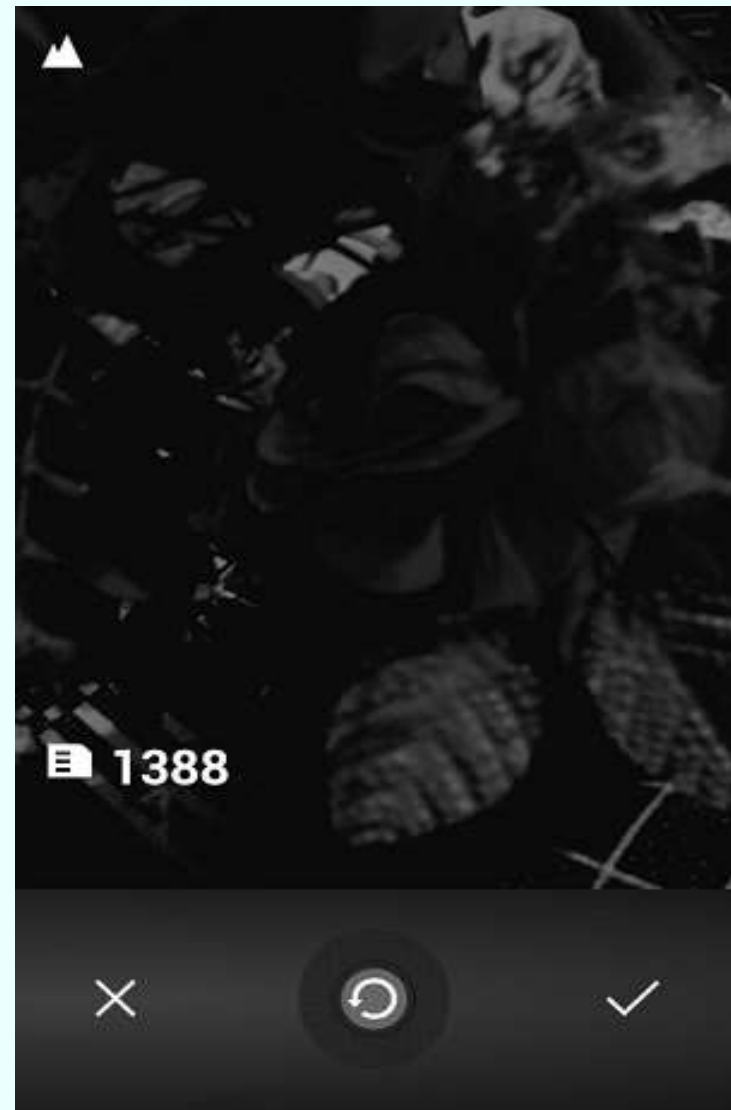
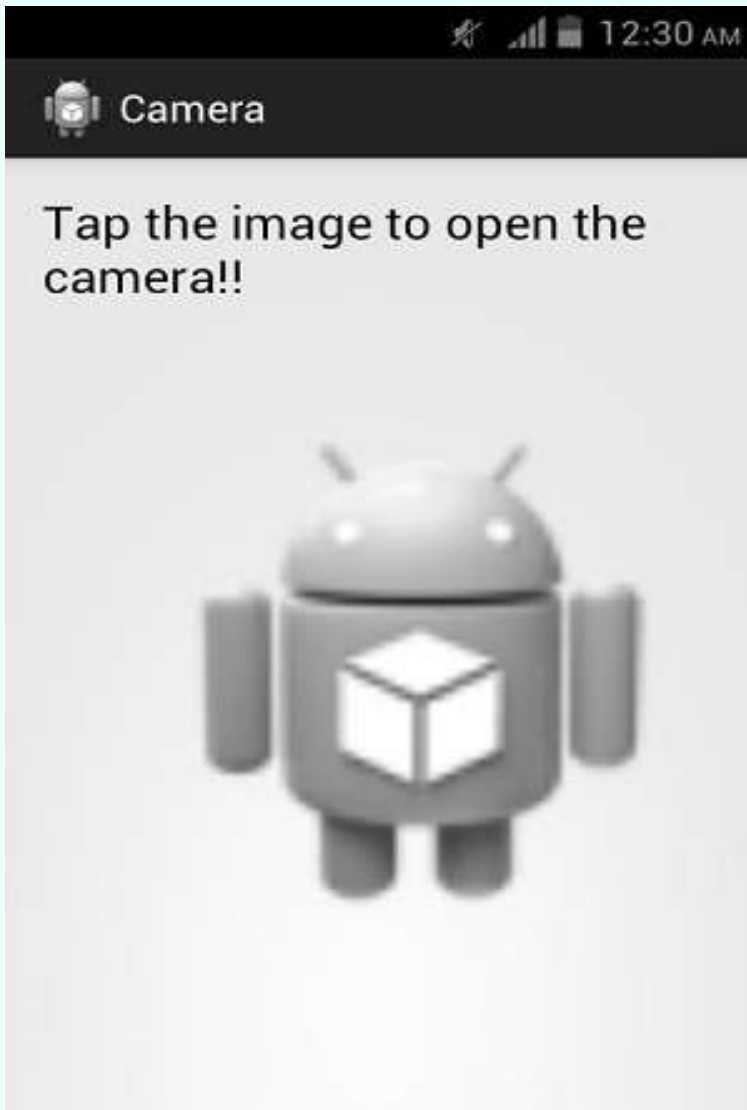
Example (4)

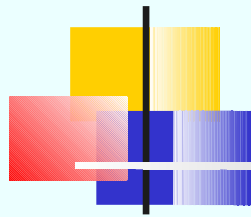
AndroidManifest.xml:

```
<application
  android:allowBackup="true"
  android:icon="@drawable/ic_launcher"
  android:label="@string/app_name"
  android:theme="@style/AppTheme" >

  <activity
    android:name="com.example.camera.MainActivity"
    android:label="@string/app_name" >
    <intent-filter>
      <action android:name="android.intent.action.MAIN" />
      <category android:name="android.intent.category.LAUNCHER"/>
    </intent-filter>
  </activity>
</application>
```

Example (5)

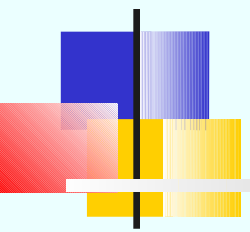




Exercises

Image sharing:

- Take a photograph and send it to other phone via telephone network
- Send it to other devices in the range of bluetooth



Questions?
